

Integrated Structures and Boundary Spectra in Sparse Prime Codings

Quantitative Orthogonality, Exact Correlation Inversion,
Prime-Gap Spectral Models, and Gumbel Boundary Data

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Audited integrated manuscript, Version 3.0 (English)

2 September 2026

Scope and provenance. This is the post-Version-2.0 audited revision of the English reconstruction of the 33-page Korean manuscript *Integrated Structures in Sparse Prime Codings*. It preserves the earlier results, corrects the anchored wheel-state congruence in the Gilbreath section, and incorporates only later deductions for which a complete argument or a precise literature input is supplied. The main additions are a quantitative prime-block transfer theorem for Möbius and Liouville sums, exact all-lag inversion of the signed prime-pair correlation, a critical rank-one model for the alternating prime-index spectrum, and the Mellin natural boundary of the dyadic numerator series, whose inverse-logarithmic coefficients are the moments of the standard Gumbel law. Exact reformulations of open problems are explicitly separated from progress on them. No claim of historical priority is made without a dedicated bibliographic search.

Abstract

Let $p_1 = 2 < p_2 < \dots$ be the primes, $g_k = p_{k+1} - p_k$, and $b_n = \pi(n) \bmod 2$. The adjacent XOR identity

$$\mathbf{1}_{\mathbb{P}}(n) = b_{n-1} \oplus b_n$$

transports the cumulative prime-parity word, without loss, among symbolic dynamics, signed correlations, doubling orbits, prime-gap continued fractions, and dyadic numerator recurrences. We make the factor-complexity bounds explicit:

$$\exp\left(\left(\frac{1}{16} - o(1)\right)(\log m)^2\right) \leq p_b(m) \leq \exp\left(C \frac{m}{\log m}\right),$$

where the lower constant is optimal within the present Maynard–CRT shadow-counting method. The same comparison and lower bound hold for every modular prime clock $\pi(n) \bmod d$. Consequently these words have zero topological entropy but are not morphic; their complex generating functions have the unit circle as a natural boundary, and the associated formal series over $\mathbb{F}_2$ is transcendental.

Every bounded sequence of vanishing Cesàro total variation is orthogonal, in Cesàro mean, to both the Möbius and Liouville functions. For arbitrary bounded prime-block labels this is sharpened to the uniform rate

$$\sum_{n \leq x} \mu(n) a_{\pi(n)}, \quad \sum_{n \leq x} \lambda(n) a_{\pi(n)} \ll \|a\|_{\infty} x \frac{\log_3 x}{\log_2 x},$$

as well as an absolute total-variation estimate for the Mertens and Liouville sums over individual prime blocks. Fixed-lag correlations yield a transition hierarchy and a prime-rooted two-wall local limit. The finite diffraction measure recovers $\pi(N)$ exactly through a chordal Wasserstein distance; after the Bragg peak is removed, the renormalized high-pass diffraction converges to Haar measure. For the signed prime-pair correlation $C_X(h)$ a discrete Green inversion recovers every overlap count and, in particular, the global parity sum at even prime endpoints. Phase tomography separately recovers all counts classified by prime-index distance. These are exact decoding identities, not solutions of the twin-prime or global parity problems.

The gap g_k is also decoded by the depth of a doubling orbit, the first partial quotient of a Gauss orbit, and a continuant denominator ratio. We prove $\log q_n = n \log \log n + O(n)$, irrationality exponent 2 for three prime-gap continued fractions, and the Hausdorff-dimension bound $2/3$ for the closure of the Gauss tails. The fixed-depth Gilbreath obstruction language is described by affine and transfer systems. A counterexample to the Version 2.0 fixed-start wheel state is given and the state is corrected by recording the actual endpoint residues; the resulting anchored right congruence is exact.

For the dyadic numerators

$$N_n = \sum_{j=1}^n (-1)^{j+1} 2^{p_n - p_j},$$

one integer N_n autonomously recovers the entire preceding prime history, and (N_n) is not P -recursive. The three lacunary support series F_Q, F_H, F_N are hypertranscendental; their nonzero algebraic specializations are transcendental and their rational specializations are Liouville numbers. The Dirichlet series $Z_N(s) = \sum N_n^{-s}$ has the imaginary axis as a natural boundary, and as $s \downarrow 0$ its full inverse-logarithmic expansion has coefficients $(-1)^m \Gamma^{(m)}(1)$, the moments of the standard Gumbel law. Hence its boundary data recovers γ and all positive-integer zeta cumulants.

Finally, the alternating prime-index Dirichlet series P_{alt} is the relative trace of a critical rank-one perturbation that changes the inverse odd-prime spectrum into the inverse even-prime spectrum. Its Jessen functions give the vague zero-measure limit on $\Re s > 1/2$; the support has exact right edge

$$\sigma_e = 1.779544653546994\dots, \quad \mathfrak{P}(\sigma_e) = 2^{1-\sigma_e},$$

and that boundary line is zero-free. We close by isolating what remains open: in particular $\sum_{n \leq x} (-1)^{\pi(n)} = o(x)$ and Erdős's alternating prime series are not proved here.

Keywords. Prime parity; factor complexity; Möbius disjointness; Liouville function; natural boundary; signed correlation; diffraction; rank-one perturbation; Gumbel moments; bounded prime gaps; continued fractions; Gilbreath conjecture; automaticity; almost-periodic Dirichlet series.

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1 Introduction and theorem map

Sparse arithmetic data often admit many encodings: an infinite word, a real number, a power series, an orbit, a continued fraction, or a difference table. The main difficulty is not constructing these encodings but determining which information survives the translation and where genuinely new input is required. This paper performs that audit for the cumulative parity of the prime-counting function.

Write

$$p_1 = 2 < p_2 < \cdots, \quad g_k = p_{k+1} - p_k,$$

and define

$$\chi_n := \mathbf{1}_{\mathbb{P}}(n), \quad b_0 := 0, \quad b_n := \pi(n) \pmod{2}, \quad \sigma_n := (-1)^{b_n}. \quad (1)$$

The organizing identity is

$$\chi_n = b_{n-1} \oplus b_n. \quad (2)$$

It is elementary, but it makes the translation dictionary exact. The word b and its coefficient sequence are recorded in OEIS; our concern is the network of consequences, not priority for the underlying digits.

The main additions in this post-Version-2.0 revision are as follows.

- (i) The Maynard–CRT complexity amplification is made explicit with constant $1/16$, optimal for the present shadow-counting argument, and is extended to the modular prime clocks $\pi(n) \bmod d$.
- (ii) The qualitative sparse-transition principle is complemented by a uniform $x \log_3 x / \log_2 x$ Möbius–Liouville bound for arbitrary prime-block labels and by an absolute block-variation theorem.
- (iii) Prime-rooted wall limits, reciprocal occurrence laws, exact diffraction recovery, a renormalized Haar limit, all-lag Green inversion, and phase tomography refine the correlation dictionary without asserting prime-pair infinitude or global parity cancellation.
- (iv) The Gauss-tail closure is proved to have Hausdorff dimension at most $2/3$. The fixed-start wheel state in the Gilbreath language is corrected by adjoining actual endpoint residues, and a counterexample explains why the earlier relative mask was insufficient.
- (v) A single dyadic numerator N_n is shown to recover the full preceding prime history autonomously; OEIS A122150 is not P -recursive. The three support series are strengthened to hypertranscendence, with arithmetic Picard fibres for F_N .
- (vi) The new Dirichlet series $Z_N(s) = \sum N_n^{-s}$ has natural boundary $\Re s = 0$ and a complete Gumbel-moment boundary expansion. The Liouville values $F_N(r)$ also yield algebraic-independence statements with π , the even zeta values, and infinitely many of those boundary moments.
- (vii) The odd and even inverse-prime spectra are connected by a critical rank-one perturbation whose relative trace is $\mathcal{P}_{\text{alt}}$. Generalized Jessen theory then gives the limiting zero measure throughout $\Re s > 1/2$.
- (viii) The global parity frontier is sharpened by exact correlation reconstruction, large-gap excursions, and convergence at every nonzero Fourier boundary frequency; the unweighted zero-frequency problem remains explicitly open.

Five corrections to the earlier version are built into the notation and statements. We write $2C_{PP}$ literally rather than introducing a tilde; the Gauss-tail indexing is $G^k(\alpha_P)$, not $G^{k+1}(\alpha_P)$; the first exponent of each of F_H and F_N has multiplicity two; and previously suggested positive-density claims for $\pi(n) \bmod 2$ are replaced by the stronger published results of Chung–Li and Alboiu. In addition, the fixed-start Gilbreath quotient now records actual endpoint residues; the former relative mask survives only for the distinct existential-translate problem.

We use the following provenance tags only as reading aids:

[E]	exact or elementary argument given here;
[A]	standard analytic theorem, with the input identified;
[L]	substantial theorem imported from the literature;
[C]	conditional statement, with its hypothesis displayed.

The tags describe proof dependence, not novelty or journal value.

2 Prime-parity coding and the doubled constant

2.1 Two finite differences

Set $a_n = b_n - b_{n-1}$ as an integer difference. Then

$$a_n = \begin{cases} (-1)^{k+1}, & n = p_k, \\ 0, & n \notin \mathbb{P}. \end{cases} \quad (3)$$

Consequently, define

$$B(z) := \sum_{n \geq 1} b_n z^n, \quad A(z) := \sum_{k \geq 1} (-1)^{k+1} z^{p_k},$$

and let $\overline{B}$ denote the coefficientwise reduction of B modulo 2. We have the exact identities

$$A(z) = (1 - z)B(z), \quad (1 + z)\overline{B}(z) = \sum_{p \in \mathbb{P}} z^p \quad \text{in } \mathbb{F}_2[[z]]. \quad (4)$$

The first identity controls complex-analytic and correlation questions; the second transports automaticity and formal algebraicity.

2.2 The constant, its digits, and the doubling detector

Define the parity prime constant

$$C_{PP} := \sum_{k \geq 1} (-1)^{k+1} 2^{-p_k}. \quad (5)$$

Evaluating the first identity in (4) at $z = 1/2$ gives

$$2C_{PP} = \sum_{n \geq 1} b_n 2^{-n} = 0.b_1 b_2 b_3 \cdots 2. \quad (6)$$

Thus $C_{PP} = 0.0b_1 b_2 \cdots 2$. This one-place shift matters when comparing the constant with OEIS entries A071986, A122150 and A122153: A071986 supplies the binary digits of $2C_{PP}$, rather than of C_{PP} under the standard radix-point convention.

Corollary 2.1 (Irrationality and non-normality). [E] *Both C_{PP} and $2C_{PP}$ are irrational and non-normal in base 2.*

Proof. If b were eventually periodic, then its adjacent XOR word would be eventually periodic. The latter is impossible because the primes are infinite and have arbitrarily long gaps. Hence the binary expansion in (6) is not eventually periodic, proving irrationality. The complexity upper bound in theorem 3.3 below is $p_b(m) = \exp(O(m/\log m)) = o(2^m)$, so for all large m at least one binary block of length m never occurs. A base-2 normal number contains every such block. Adding or removing the initial zero digit does not change this conclusion for C_{PP} . $\square$

Let $T(x) = 2x \pmod{1}$ and $R = [1/4, 3/4)$. Reading the first two digits after n shifts shows that, for every $n \geq 0$,

$$\mathbf{1}_R(T^n(2C_{PP})) = \chi_{n+2}. \quad (7)$$

The visit times to R are therefore $p_k - 2$, and successive return times are precisely the gaps g_k . This is an exact coordinate change, not a new primality test.

3 Factor complexity, morphisms, automata, and natural boundaries

Let $p_{\mathbb{P}}(m)$ and $p_b(m)$ denote the number of distinct length- m factors of the prime characteristic word and of $b = b_0b_1\cdots$, respectively.

Theorem 3.1 (Exact comparison). [E] *For every $m \geq 1$,*

$$p_{\mathbb{P}}(m) \leq p_b(m+1) \leq 2p_{\mathbb{P}}(m). \quad (8)$$

Proof. Adjacent XOR sends every factor of b of length $m+1$ to a unique prime-indicator factor of length m . Conversely, an indicator factor and the initial bit of a lift determine all remaining bits by cumulative XOR. There are at most two choices for that initial bit. $\square$

3.1 A sieve upper bound

We use the following standard uniform form of the one-dimensional Selberg upper-bound sieve; see, for example, [15, Chapter 6].

Lemma 3.2 (Uniform omitted-residue sieve). [A] *Put $z = m^{1/100}$. For each prime $\ell \leq z$ choose an arbitrary residue $r_\ell \pmod{\ell}$, and set*

$$\mathcal{R}(\mathbf{r}) := \{1 \leq j \leq m : j \not\equiv r_\ell \pmod{\ell} \text{ for every } \ell \leq z\}.$$

Uniformly in the residue choices,

$$|\mathcal{R}(\mathbf{r})| \ll \frac{m}{\log m}.$$

Proof. The local density removed at ℓ is $1/\ell$, independently of the selected residue. The dimension-one Selberg upper-bound sieve at level z therefore gives

$$|\mathcal{R}(\mathbf{r})| \ll m \prod_{\ell \leq z} \left(1 - \frac{1}{\ell}\right) \asymp \frac{m}{\log z} \asymp \frac{m}{\log m}.$$

The Chinese remainder theorem makes the remainder estimates uniform in $\mathbf{r}$. $\square$

Theorem 3.3 (Subexponential upper bound). [A] *There is an absolute constant $C > 0$ such that*

$$p_{\mathbb{P}}(m), p_b(m) \leq \exp\left(C \frac{m}{\log m}\right). \quad (9)$$

Proof. Consider a length- m prime-indicator block starting at $N > z$. For every $\ell \leq z$, its prime positions omit at least one residue class modulo ℓ : the unique position congruent to $-N \pmod{\ell}$ would otherwise represent a multiple of ℓ larger than ℓ . Hence the set of 1-positions lies in some $\mathcal{R}(\mathbf{r})$. There are

$$\prod_{\ell \leq z} \ell = \exp(\vartheta(z)) = \exp(O(z))$$

possible residue vectors and, for each, at most $2^{|\mathcal{R}(\mathbf{r})|}$ subsets. By lemma 3.2 and $z = o(m/\log m)$, the product is $\exp(O(m/\log m))$. Starts $N \leq z$ contribute only $O(z)$ further blocks. Transfer to b by theorem 3.1. $\square$

3.2 Maynard–CRT amplification

The upper bound has an unconditional superpolynomial counterpart. The construction is included because the CRT admissibility check is essential.

Theorem 3.4 (Explicit complexity amplification). [L+A] *As $L \rightarrow \infty$,*

$$\log p_{\mathbb{P}}(L), \log p_b(L) \geq \left(\frac{1}{16} - o(1)\right) (\log L)^2. \quad (10)$$

The constant $1/16$ is optimal within the present Maynard–shadow-counting argument.

Proof. Let T be the set of primes in $(L/2, L]$ and write $K = |T| \asymp L/\log L$. For large L we have $K < L/2$. The offset set T is admissible: if $\ell \leq K$ is prime, the residue 0 is absent because every member of T exceeds ℓ ; if $\ell > K$, then K offsets cannot cover all ℓ residue classes.

Set $k = \lfloor \sqrt{K} \rfloor$. For every k -element subset $U \subset T$, choose distinct primes $q_a > L$ for all $a \in [1, L] \setminus U$. The Chinese remainder theorem gives N_U with

$$N_U \equiv -a \pmod{q_a} \quad (a \in [1, L] \setminus U),$$

and put $M_U = \prod_{a \notin U} q_a$. Consider the common-slope linear forms

$$\mathcal{L}_u(t) := M_U t + N_U + u \quad (u \in U).$$

They remain admissible. Indeed, if a prime q divides M_U , then $q = q_a$ for one $a \notin U$ and $N_U + u \equiv u - a \not\equiv 0 \pmod{q}$, because $0 < |u - a| < L < q$. If $q \nmid M_U$, multiplication by M_U^{-1} preserves the number of occupied root classes, and admissibility follows from that of $U \subset T$.

Maynard’s variational estimate $M_k > \log k - 2 \log \log k - 2$ and his criterion $M_k > 4m$ [16] show that, for every U and infinitely many t , at least

$$r = \left(\frac{1}{4} - o(1)\right) \log k$$

of the values $\mathcal{L}_u(t)$ are prime. Taking t large also ensures that every $N_U + M_U t + a$ with $a \notin U$ is a composite multiple of q_a . Thus each U produces a length- L prime pattern $S \subset U$ with $|S| \geq r$.

A fixed pattern S of size at least r can arise from at most $\binom{K-r}{k-r}$ choices of U . Consequently the number of distinct patterns is at least

$$\frac{\binom{K}{k}}{\binom{K-r}{k-r}} = \prod_{j=0}^{r-1} \frac{K-j}{k-j} \geq \exp\left(\left(\frac{1}{16} - o(1)\right)(\log K)^2\right) = \exp\left(\left(\frac{1}{16} - o(1)\right)(\log L)^2\right).$$

The bound for b follows from (8), after an immaterial shift of L .

More generally, if $k = K^\alpha$ with $0 < \alpha < 1$, the same computation gives the constant $\frac{1}{4}\alpha(1-\alpha)$. Its maximum is $1/16$, attained at $\alpha = 1/2$. $\square$

Corollary 3.5 (Modular prime clocks). [E+A] *For every integer $d \geq 2$, let*

$$c_n^{(d)} := \pi(n) \pmod{d}$$

and denote its factor complexity by $p_d(m)$. Then

$$p_{\mathbb{P}}(m) \leq p_d(m+1) \leq d p_{\mathbb{P}}(m), \quad (11)$$

so

$$\log p_d(m) \geq \left(\frac{1}{16} - o(1)\right)(\log m)^2, \quad \log p_d(m) = O(m/\log m).$$

Every invariant measure on the shift orbit closure of $c^{(d)}$ is supported on constant fixed points.

Proof. Taking adjacent differences modulo d recovers the prime indicator, and an indicator factor together with its first clock value has at most d lifts. This proves (11). The changes of $c^{(d)}$ occur only at primes, whose upper Banach density is zero; the orbit-closure argument of [remark 4.2](#) therefore forces every invariant measure onto constant points. The assertion does not claim that every one of the d constant points belongs to the orbit closure. $\square$

Corollary 3.6 (Non-morphicity). [A] *Neither the prime characteristic word nor the cumulative parity word b is morphic.*

Proof. Every morphic word has factor complexity $O(n^2)$ by the general complexity theorem of Devyatov [10]. The lower bound (10) exceeds every fixed power of L . $\square$

Theorem 3.7 (Conditional optimal order). [C] *Assume the prime k -tuple conjecture for every finite admissible tuple. Then*

$$p_{\mathbb{P}}(L), p_b(L) = \exp\left(\Theta\left(\frac{L}{\log L}\right)\right).$$

Proof. Use the same admissible T . For each prescribed subset $S \subset T$, force every position in $[1, L] \setminus S$ composite by distinct moduli as above. The remaining forms indexed by S are admissible, so the conjecture makes all of them prime simultaneously. This realizes $2^K = \exp(\Omega(L/\log L))$ distinct patterns. The reverse inequality is [theorem 3.3](#). $\square$

3.3 Entropy, automaticity, and two natural boundaries

Corollary 3.8. [A] *The word b is aperiodic and*

$$m + 1 \leq p_b(m) \leq \exp(O(m/\log m)), \quad h_{\text{top}}(b) = 0.$$

Proof. Aperiodicity follows from (2), the infinitude of primes, and arbitrarily long prime gaps. Morse–Hedlund gives $p_b(m) \geq m + 1$; the upper bound gives zero entropy. $\square$

Theorem 3.9 (Automaticity and formal transcendence). [L+E] *For no integer base $q \geq 2$ is b a q -automatic word. Moreover*

$$\overline{B}(z) = \sum_{n \geq 1} b_n z^n \in \mathbb{F}_2[[z]]$$

is transcendental over $\mathbb{F}_2(z)$.

Proof. Automatic words are closed under shifts and coordinatewise finite-alphabet operations. If b were q -automatic, (2) would make the prime characteristic word q -automatic, contradicting the nonrecognizability theorem for primes [13, 2]. For $q = 2$, Christol’s theorem [7] identifies algebraic power series over $\mathbb{F}_2(z)$ with 2-automatic coefficient sequences, proving the second assertion. $\square$

Theorem 3.10 (Natural boundaries). [A] *The functions $A(z)$ and $B(z)$ have radius of convergence 1 and the unit circle as a natural boundary. Hence both are transcendental over $\mathbb{C}(z)$ and are not D-finite.*

Proof. Their coefficients lie in finite sets of integers and have infinitely many nonzero terms, so the radius is 1. A rational power series with coefficients in a finite set is eventually periodic. Rationality of B would make b eventually periodic; rationality of A would make its signed prime support eventually periodic. Both are impossible. The Pólya–Carlson dichotomy [21, 5] therefore gives the natural boundary. An algebraic function, and more generally a D-finite function, has only finitely many singularities in the finite plane, so neither class can contain A or B . $\square$

Remark 3.11. Formal or functional transcendence does not imply the transcendence of the particular value $B(1/2) = 2C_{PP}$. The latter remains open.

4 Sparse transitions and Möbius–Liouville orthogonality

The prime-parity word has zero entropy for a quantitative combinatorial reason, but it also has a much stronger measure-theoretic degeneracy: it changes only at a zero-density set. The following criterion extracts the multiplicative consequence. The deep input is the measure-complexity theorem of Huang, Wang and Ye [14]; the deduction below is short and should be read as an application of that theorem. Wei’s work on asymptotically periodic functions [26] is closely related; in particular it covers block-constant sequences whose successive block lengths tend to infinity. Our density criterion permits recurring short clusters and therefore applies directly to prime boundaries.

For a bounded complex sequence $u = (u_n)_{n \geq 1}$ put

$$V_u(N) := \sum_{n=1}^N |u_{n+1} - u_n|.$$

Theorem 4.1 (Vanishing Cesàro variation). [L+E] *If u is bounded and $V_u(N) = o(N)$, then*

$$\sum_{n \leq N} \mu(n)u_n = o(N), \quad \sum_{n \leq N} \lambda(n)u_n = o(N). \quad (12)$$

Here μ and λ denote the Möbius and Liouville functions.

Proof. Let $K \subset \mathbb{C}$ be the compact closure of the values of u . Extend u by a constant negative tail to a point $x \in K^{\mathbb{Z}}$, let S be the bilateral shift, and take the orbit closure X of x . Given any sequence $N_j \rightarrow \infty$, pass to a subsequence for which

$$\rho_j := \frac{1}{N_j} \sum_{n=1}^{N_j} \delta_{S^n x} \implies \rho$$

weakly. The limit is S -invariant. For the continuous function $\Psi(y) = |y_1 - y_0|$,

$$\int_X \Psi \, d\rho = \lim_j \frac{V_u(N_j)}{N_j} = 0.$$

Thus $y_1 = y_0$ for ρ -almost every y ; invariance and a countable intersection show that ρ is supported on constant sequences. The shift is therefore the identity ρ -almost everywhere. Such a measure has bounded, hence subpolynomial, measure complexity. Huang–Wang–Ye’s pointwise criterion [14, Theorem 1.1’], applied to the coordinate observable $y \mapsto y_0$, gives

$$\frac{1}{N_j} \sum_{n \leq N_j} \mu(n)u_n \rightarrow 0$$

along this subsequence. A bad-subsequence contradiction proves the full Möbius limit.

For Liouville, use the exact square-divisor identity

$$\lambda(n) = \sum_{d^2 | n} \mu(n/d^2).$$

For each fixed d , the dilated sequence $u_m^{(d)} = u_{d^2 m}$ also has vanishing Cesàro variation, since

$$\sum_{m < H} |u_{d^2(m+1)} - u_{d^2 m}| \leq \sum_{r \leq d^2 H + O(d^2)} |u_{r+1} - u_r| = o(H).$$

Hence the contribution of every fixed d is $o_d(N/d^2)$. For fixed D , the terms $d \leq D$ total $o(N)$, while the remaining terms are bounded by

$$\|u\|_{\infty} N \sum_{d > D} d^{-2} = O(\|u\|_{\infty} N/D).$$

Let $N \rightarrow \infty$ and then $D \rightarrow \infty$. □

Remark 4.2 (Orbit-closure form). If the stronger uniform variation condition

$$\lim_{H \rightarrow \infty} \sup_m \frac{1}{H} \sum_{m \leq n < m+H} |u_{n+1} - u_n| = 0$$

holds, then every invariant measure on the entire orbit closure is supported on constant fixed points. Huang–Wang–Ye’s system-wide theorem [14, Theorem 1.2] gives Möbius disjointness for the whole subshift. For a finite-valued word it is enough that its change set have upper Banach density zero. Ordinary density zero suffices for the pointwise [theorem 4.1](#), but not for this orbit-closure statement.

The primes themselves have upper Banach density zero. Indeed, for $Q_z = \prod_{p \leq z} p$, primes larger than z occupy only the $\varphi(Q_z)$ reduced classes modulo Q_z ; uniformly over intervals of length H their upper density is at most $\varphi(Q_z)/Q_z + O_z(1/H)$, which tends to zero as first H and then z tend to infinity.

Corollary 4.3 (Arbitrary labels on prime-gap blocks). [L+E] *For every bounded complex sequence $(a_k)_{k \geq 0}$,*

$$\sum_{n \leq N} \mu(n) a_{\pi(n)} = o(N), \quad \sum_{n \leq N} \lambda(n) a_{\pi(n)} = o(N). \quad (13)$$

The same is true with $\pi(n-1)$ in place of $\pi(n)$.

Proof. The sequence $a_{\pi(n-1)}$ changes only at primes. Its total variation is at most $2 \sup_k |a_k| \pi(N) = o(N)$, so [theorem 4.1](#) applies. Replacing $\pi(n-1)$ by $\pi(n)$ changes only the prime-indexed terms, whose total contribution is $O(\pi(N)) = o(N)$. $\square$

Taking $a_k = (-1)^k$ yields, unconditionally,

$$\sum_{n \leq N} \mu(n) (-1)^{\pi(n)} = o(N), \quad \sum_{n \leq N} \lambda(n) (-1)^{\pi(n)} = o(N). \quad (14)$$

Every fixed finite-window coding $\Phi(\sigma_{n+h_1}, \dots, \sigma_{n+h_r})$, and in particular every finite shift product, changes only on a finite union of shifts of the primes. It therefore satisfies the same two orthogonality conclusions.

Two illustrative applications require no information about the underlying prime race. If $\pi_2(n)$ counts twin-prime pairs with lower endpoint at most n , then $(-1)^{\pi_2(n)}$ changes on a subset of the primes; both its μ - and λ -correlations are $o(N)$ whether the twin primes are finite or infinite. Similarly,

$$\operatorname{sgn}(\pi(n; 4, 3) - \pi(n; 4, 1))$$

changes only at primes, so it too is orthogonal to both multiplicative functions. These are sparse-boundary consequences, not results on twin-prime infinitude or the limiting Chebyshev bias.

4.1 A quantitative prime-block transfer theorem

The qualitative dynamical argument above is uniform in the labels but supplies no rate. Short-interval cancellation for multiplicative functions gives a complementary quantitative statement. We write $\log_j$ for the j -fold iterated logarithm.

Theorem 4.4 (Quantitative prime-block transfer). [L+E] *Uniformly for every bounded complex sequence (a_k) ,*

$$\sum_{n \leq x} \mu(n) a_{\pi(n)}, \quad \sum_{n \leq x} \lambda(n) a_{\pi(n)} \ll \|a\|_{\infty} x \frac{\log_3 x}{\log_2 x}. \quad (15)$$

If f is either μ or λ and $F(y) = \sum_{n \leq y} f(n)$, then

$$\sum_{p_{k+1} \leq x} |F(p_{k+1} - 1) - F(p_k - 1)| \ll x \frac{\log_3 x}{\log_2 x}. \quad (16)$$

The same bounds hold for $\pi(n-1)$, fixed finite-window codings, and fixed finite shift products, with constants depending on the window.

Proof. It is enough to work on a dyadic interval $[X, 2X]$. Put

$$H = \left\lfloor \frac{\log X}{(\log_2 X)^4} \right\rfloor, \quad G = \left\lfloor \frac{\log X}{(\log_2 X)^2} \right\rfloor, \quad W_H(t) = \sum_{t < n \leq t+H} f(n).$$

The total length of prime-gap blocks of length at most G is $O(XG/\log X)$. If a block I has length greater than G , sliding an H -window through its interior gives

$$\left| \sum_{n \in I} f(n) \right| \leq \frac{1}{H} \sum_{[t+1, t+H] \subset I} |W_H(t)| + O(H). \quad (17)$$

There are $O(X/G)$ long blocks, so their total endpoint error is $O(XH/G)$.

The mean-square short-interval estimate of Matomäki–Radziwiłł–Tao and Cauchy–Schwarz give

$$\frac{1}{H} \sum_{X \leq t \leq 2X} |W_H(t)| \ll X \left(\sqrt{e^{-M(f; X)} M(f; X)} + \frac{\log \log H}{\log H} + \frac{1}{(\log X)^{1/100}} \right),$$

where $M(f; X)$ is the pretentious-distance parameter in [38, Appendix A, Theorem A.1]. Since $\mu(p) = \lambda(p) = -1$, the bound [38, (1.12)] gives

$$M(\mu; X), M(\lambda; X) \geq \left(\frac{1}{3} - o(1) \right) \log_2 X.$$

Thus the first and third terms are $O((\log X)^{-c})$ for some absolute $c > 0$, whereas $\log H \sim \log_2 X$ and $\log \log H \sim \log_3 X$. Hence the preceding display is $\ll X \log_3 X / \log_2 X$. Combining this with the short-block contribution and (17) proves the same bound for the total absolute sum of the multiplicative block sums. Multiplication by arbitrary labels of modulus at most $\|a\|_\infty$ proves (15); choosing on every block the conjugate phase of its sum proves (16). Dyadic summation completes the proof. A fixed window changes only $O(1)$ neighboring block boundaries, so the final assertion follows by the same decomposition. $\square$

4.2 Weighted Mertens sums along zero-density sequences

Let

$$M(x) := \sum_{n \leq x} \mu(n), \quad L(x) := \sum_{n \leq x} \lambda(n).$$

Corollary 4.5 (Sparse sampling theorem). [L+E] *Let $1 \leq t_1 < t_2 < \dots$ be infinite with $\#\{k : t_k \leq x\} = o(x)$. If (c_k) has uniformly bounded partial sums $C_K = \sum_{k \leq K} c_k$, then*

$$\sum_{k \leq K} c_k M(t_k) = o(t_K), \quad \sum_{k \leq K} c_k L(t_k) = o(t_K). \quad (18)$$

Proof. Set $a_0 = 0$, $a_k = -C_k$, and

$$u_n = a_{A(n)}, \quad A(n) := \#\{j : t_j < n\}.$$

The word u is bounded and changes only at the zero-density set $\{t_k\}$. At $N_K = t_K + 1$, discrete summation by parts gives the exact identities

$$\sum_{n \leq N_K} \mu(n) u_n = M(N_K) a_K + \sum_{k \leq K} c_k M(t_k),$$

$$\sum_{n \leq N_K} \lambda(n) u_n = L(N_K) a_K + \sum_{k \leq K} c_k L(t_k).$$

The left sides are $o(N_K)$ by [theorem 4.1](#); also $M(N) = o(N)$ by the prime number theorem, while $L(N) = o(N)$ follows from the Liouville part of that theorem with $u \equiv 1$. Since a_K is bounded and $K = o(t_K)$, the result follows. $\square$

In particular, for every zero-density increasing sequence,

$$\sum_{k \leq K} (-1)^k M(t_k) = o(t_K), \quad \sum_{k \leq K} (-1)^k L(t_k) = o(t_K). \quad (19)$$

This includes $t_k = p_k$. More generally all mean-zero periodic weights and all nonprincipal periodic characters are covered.

There is also an absolute block-cancellation consequence. Choose on each interval $(t_k, t_{k+1}]$ the phase conjugate to the corresponding Möbius block sum and apply [theorem 4.1](#) to the resulting bounded block label. Then

$$\sum_{k < K} |M(t_{k+1}) - M(t_k)| = o(t_K), \quad (20)$$

and analogously for L . For prime blocks, [theorem 4.4](#) supplies the explicit rate in (16); for a general zero-density sequence the assertion remains qualitative.

5 Fixed-lag correlations and transition strata

For fixed $h \geq 1$ define

$$D_h(N) := \#\{n \leq N : b_{n+h} \neq b_n\}, \quad \rho_N(h) := \frac{1}{N} \sum_{n \leq N} \sigma_n \sigma_{n+h}.$$

Theorem 5.1 (Fixed-lag asymptotic). [\[A\]](#) *For every fixed h ,*

$$D_h(N) = h\pi(N) + O_h\left(\frac{N}{\log^2 N}\right), \quad (21)$$

$$\rho_N(h) = 1 - \frac{2h}{\log N} + O_h(\log^{-2} N). \quad (22)$$

Proof. Put $X_h(n) = \pi(n+h) - \pi(n)$. The bits disagree precisely when $X_h(n)$ is odd, and endpoint counting gives

$$\sum_{n \leq N} X_h(n) = h\pi(N) + O_h(1).$$

On a window containing at most one prime, $X_h(n)$ equals its oddness indicator. A discrepancy requires a pair of primes among $O_h(1)$ fixed offset pairs. The Selberg prime-pair upper bound makes the number of such windows $O_h(N/\log^2 N)$, proving (21). Finally $\sigma_n \sigma_{n+h} = 1 - 2\mathbf{1}_{b_n \neq b_{n+h}}$ and the prime number theorem give (22). $\square$

For every fixed integer h , $\rho_N(h) \rightarrow 1$. Hence the Herglotz measure associated with these Besicovitch autocorrelation coefficients is δ_0 . This does not assert that the full orbit closure is uniquely ergodic; [remark 4.2](#) instead describes all of its invariant measures, which are mixtures of constant fixed points.

For a binary word $w = w_1 \cdots w_m$ define its transition count

$$\tau(w) := \#\{1 \leq j < m : w_j \neq w_{j+1}\},$$

and let $A_w(N)$ count its occurrences starting at positions at most N .

Theorem 5.2 (Transition-count hierarchy). [A] *For every fixed word w ,*

$$A_w(N) \ll_w \frac{N}{(\log N)^{\tau(w)}}. \quad (23)$$

If $\tau(w) \geq 2$, the sum of the reciprocals of its occurrence positions converges.

Proof. Each transition in w forces, through (2), a prime at a distinct fixed offset from the starting point. A fixed-tuple upper-bound sieve gives (23); a local congruence obstruction may make the actual count eventually zero. Partial summation proves reciprocal summability for $\tau(w) \geq 2$. $\square$

5.1 Prime-rooted walls and a reciprocal phase transition

The usual integer-rooted local limit is dominated by long constant blocks. Centering instead at a uniformly selected prime exposes the transition itself.

Theorem 5.3 (Prime-rooted domain-wall limit). [A+E] *Fix $R \geq 1$, choose $p_k \leq X$ uniformly among the primes, and observe*

$$(b_{p_k-R}, \dots, b_{p_k-1} \mid b_{p_k}, \dots, b_{p_k+R}).$$

Its distribution converges, with cylinder error $O_R(1/\log X)$, to

$$\frac{1}{2}\delta_{0^\infty|1^\infty} + \frac{1}{2}\delta_{1^\infty|0^\infty}. \quad (24)$$

Proof. A centered block fails to be a single wall only if another prime lies within distance R of p_k . Summing the fixed-shift prime-pair sieve bound over $1 \leq d \leq R$ shows that there are $O_R(X/\log^2 X)$ exceptional centers. Division by $\pi(X)$ gives the stated error. Odd- and even-indexed primes differ in number by at most one and give the two opposite orientations. $\square$

For a finite word w , let

$$H_w(X) := \sum_{\substack{n \leq X \\ b_n b_{n+1} \cdots b_{n+|w|-1} = w}} \frac{1}{n}.$$

Theorem 5.4 (Reciprocal three-phase law). [A+E] *There is a constant κ_w in the one-transition case such that*

$$\begin{aligned} \tau(w) = 0 : \quad & H_w(X) \asymp_w \log X, \\ \tau(w) = 1 : \quad & H_w(X) = \frac{1}{2} \log \log X + \kappa_w + O_w(1/\log X), \\ \tau(w) \geq 2 : \quad & H_w(\infty) - H_w(X) = O_w((\log X)^{1-\tau(w)}). \end{aligned} \quad (25)$$

For $\tau(w) = 1$ one has more precisely

$$A_w(X) = \frac{1}{2}\pi(X) + O_w(X/\log^2 X). \quad (26)$$

Proof. If $\tau(w) = 1$, its unique transition specifies one prime at a fixed offset and one of the two prime-index parities. These parities split the first $\pi(X)$ primes equally up to one; an additional prime inside the fixed window accounts for $O_w(X/\log^2 X)$ exceptions. This proves (26), and partial summation together with the Mertens theorem for reciprocal primes gives the middle line of (25). For $\tau(w) \geq 2$, apply partial summation to (23). Finally, a constant word occurs on a positive lower density of positions by (154), after discarding the zero-density set lying within a fixed distance of a prime; the trivial upper bound completes the first line. $\square$

5.2 Finite diffraction and its first high-pass scale

Put

$$S_N(\theta) := \sum_{n=1}^N \sigma_n e^{in\theta}, \quad d\hat{\mu}_N(\theta) := \frac{|S_N(\theta)|^2}{N} \frac{d\theta}{2\pi}.$$

This is a probability measure on the circle. Write $d_{\text{ch}}(e^{i\theta}, 1) = |e^{i\theta} - 1|$.

Proposition 5.5 (Exact prime count from finite diffraction). [E]

$$W_{2,\text{ch}}(\hat{\mu}_N, \delta_0)^2 = \frac{4\pi(N) + 2}{N}, \quad \pi(N) = \frac{N}{4} W_{2,\text{ch}}(\hat{\mu}_N, \delta_0)^2 - \frac{1}{2}. \quad (27)$$

Consequently the prime number theorem is equivalent to

$$\sqrt{\log N} W_{2,\text{ch}}(\hat{\mu}_N, \delta_0) \longrightarrow 2.$$

Proof. The only coupling with the point mass δ_0 gives

$$W_{2,\text{ch}}^2 = 2 - 2\Re \widehat{\hat{\mu}_N}(1).$$

Parseval and the adjacent sign change at every prime give

$$\widehat{\hat{\mu}_N}(1) = \frac{1}{N} \sum_{n < N} \sigma_n \sigma_{n+1} = \frac{N - 1 - 2\pi(N)}{N},$$

which is (27). $\square$

Theorem 5.6 (Renormalized high-pass diffraction). [A+E] Define

$$d\hat{\nu}_N(\theta) := (\log N) \sin^2 \frac{\theta}{2} d\hat{\mu}_N(\theta).$$

Then

$$\hat{\nu}_N \Longrightarrow \frac{d\theta}{2\pi}. \quad (28)$$

Thus the unfiltered diffraction converges to the Bragg mass δ_0 , whereas its first nontrivial high-pass scale is Haar measure.

Proof. For fixed $h \in \mathbb{Z}$, put

$$r_N(h) = \frac{1}{N} \sum_{n=1}^{N-|h|} \sigma_n \sigma_{n+|h|}.$$

By theorem 5.1, $r_N(h) = 1 - 2|h|/\log N + O_h(\log^{-2} N)$. Hence

$$\widehat{\hat{\nu}_N}(h) = \log N \left(\frac{1}{2} r_N(h) - \frac{1}{4} r_N(h-1) - \frac{1}{4} r_N(h+1) \right).$$

The discrete second difference of $|h|$ is 2 at $h = 0$ and zero elsewhere. The displayed Fourier coefficient therefore tends to 1 at $h = 0$ and to 0 otherwise. Positivity and Fourier uniqueness on the circle prove (28). $\square$

Corollary 5.7 (Pure diffraction without Besicovitch phase coherence). [L+E] *With respect to the prefix Følner sequence $[1, N]$, the prime-parity sign comb has pure-point diffraction δ_0 , but (σ_n) is not Besicovitch B^2 -almost periodic.*

Proof. The consistent-phase characterization of Besicovitch almost periodic diffraction [35] would force a B^2 -almost periodic sequence with the single diffraction atom of mass one to agree in B^2 with a constant phase. This is impossible here because (154) gives positive lower density to each of the values $+1$ and -1 . $\square$

6 Correlation curvature and individual prime-gap thresholds

6.1 The discrete correlation Laplacian

For a finitely supported sequence u set $(\Delta u)_n = u_n - u_{n-1}$ and $T_u(h) = \sum_n u_n u_{n+h}$.

Proposition 6.1 (Correlation curvature). [E]

$$T_{\Delta u}(h) = 2T_u(h) - T_u(h-1) - T_u(h+1). \quad (29)$$

Proof. Expand $(1 - S)$ in both factors and shift the summation indices. For truncations of an infinite sequence, the corresponding endpoint terms must be retained. $\square$

Thus the negative discrete Laplacian of a dense cumulative correlation recovers the correlation of its sparse signed difference.

6.2 All-lag Green inversion and exact parity recovery

For $X = p_{2m}$, put

$$C_X(h) := \sum_{\substack{i < j, p_j \leq X \\ p_j - p_i = h}} (-1)^{j-i}, \quad \mathcal{T}(x) := \sum_{n \leq x} (-1)^{\pi(n)},$$

and

$$D_X(h) := \sum_{n=1}^{X-1-h} b_n b_{n+h} \quad (h \geq 0),$$

with $D_X(h) = 0$ when the sum is empty.

Theorem 6.2 (All-lag Green inversion). [E] *For every $h \geq 1$,*

$$C_X(h) = 2D_X(h) - D_X(h-1) - D_X(h+1). \quad (30)$$

Conversely, for every $h \geq 0$,

$$D_X(h) = - \sum_{t>h} (t-h) C_X(t). \quad (31)$$

In particular,

$$\#\{n \leq X-1 : \pi(n) \text{ is odd}\} = - \sum_{h \geq 1} h C_X(h), \quad (32)$$

$$\mathcal{T}(X-1) = X-1 + 2 \sum_{h \geq 1} h C_X(h). \quad (33)$$

The correlation also obeys the exact sum rules

$$\sum_{h \geq 1} C_X(h) = -m, \quad \sum_{h \geq 1} h^2 C_X(h) = - \left(\sum_{h \geq 1} h C_X(h) \right)^2. \quad (34)$$

Proof. Extend $u_n = b_n \mathbf{1}_{1 \leq n \leq X-1}$ by zero to $\mathbb{Z}$. Since $X = p_{2m}$, its difference is supported exactly on $p_1, \dots, p_{2m}$ and $(\Delta u)_{p_k} = (-1)^{k+1}$. Hence $T_{\Delta u}(h) = C_X(h)$, so [proposition 6.1](#) gives (30). Twofold summation, using the finite support of D_X , gives (31); taking $h = 0$ yields (32) and (33).

Write $\epsilon_k = (-1)^{k+1}$. Since $\sum_{k \leq 2m} \epsilon_k = 0$,

$$\sum_h C_X(h) = \sum_{i < j} \epsilon_i \epsilon_j = -m, \quad \sum_h h C_X(h) = \sum_k \epsilon_k p_k = -D_X(0).$$

Expanding $(p_j - p_i)^2$ and again using $\sum \epsilon_k = 0$ gives

$$\sum_h h^2 C_X(h) = \sum_{i < j} \epsilon_i \epsilon_j (p_j - p_i)^2 = - \left(\sum_k \epsilon_k p_k \right)^2,$$

which completes (34). $\square$

Thus the open estimate $\mathcal{T}(x) = o(x)$ is, along even prime endpoints, equivalent to the single signed-moment asymptotic

$$\sum_{h \geq 1} h C_{p_{2m}}(h) = -\frac{p_{2m}}{2} + o(p_{2m}). \quad (35)$$

This identifies the required statistic but does not estimate it.

6.3 Prime-rank phase tomography

For $|z| = 1$, set

$$u_z(n) := z^{\pi(n)}, \quad d_z(n) := u_z(n) - u_z(n-1).$$

Then $d_z(p_k) = (z-1)z^{k-1}$ and $d_z(n) = 0$ off the primes. Let

$$G_{r,X}(h) := \#\{i : p_{i+r} - p_i = h, p_{i+r} \leq X\}.$$

Proposition 6.3 (Exact phase decoder). [\[E\]](#) For $z \neq 1$,

$$\mathcal{F}_{X,h}(z) := \frac{1}{|z-1|^2} \sum_{n \leq X-h} \overline{d_z(n)} d_z(n+h) = \sum_{r=1}^h G_{r,X}(h) z^r. \quad (36)$$

The value at $z = 1$ is defined by polynomial continuation. If $q > h$ and $\omega = e^{2\pi i/q}$, then

$$G_{r,X}(h) = \frac{1}{q} \sum_{a=0}^{q-1} \omega^{-ar} \mathcal{F}_{X,h}(\omega^a). \quad (37)$$

Thus all prime-index distances at a fixed additive distance are recovered exactly; in particular, $G_{1,X}(h)$ is the number of consecutive prime gaps equal to h , with no restriction $h < 2H_*$.

Proof. Every nonzero summand on the left corresponds to a prime pair $p_i, p_j = p_i + h$, and after division by $|z - 1|^2$ its value is z^{j-i} . Grouping by $r = j - i$ proves (36); finite Fourier inversion proves (37). $\square$

The decoder reconstructs the counts $G_{r,X}(h)$, not the temporal order of the gaps. Recovering ordered finite gap words requires X -increments or higher multivariate correlations.

6.4 An exact twin-prime curvature identity

Put

$$R_N(h) := \sum_{n=1}^N \sigma_n \sigma_{n+h}, \quad \Pi_2(X) := \#\{p \leq X - 2 : p, p + 2 \in \mathbb{P}\}.$$

Theorem 6.4 (Twin-prime curvature). [E]

$$R_N(3) - 2R_N(2) + R_N(1) = 4\Pi_2(N + 3) - 2 - 2\chi_{N+3}. \quad (38)$$

Proof. Use $\sigma_n \sigma_{n+h} = \prod_{j=1}^h (1 - 2\chi_{n+j})$ for $h = 1, 2, 3$. In the displayed second difference, the interior linear terms telescope and the nonadjacent quadratic term $4\chi_{n+1}\chi_{n+3}$ counts twin pairs. The only adjacent prime pair is $(2, 3)$; collecting it and the two endpoints gives (38). $\square$

Twin-prime infinitude is equivalent to unboundedness above of the left side. The theorem is an exact reformulation, not a new lower bound for prime pairs.

6.5 Cumulative and individual signed correlations

Recall the signed correlation $C_X(h)$ defined above, and put

$$L_{i,X}(H) := \#\{j > i : p_j \leq X, p_j - p_i \leq H\}.$$

Theorem 6.5 (Cumulative anti-correlation). [E]

$$\sum_{h=1}^H C_X(h) = -\#\{i : L_{i,X}(H) \text{ is odd}\} \leq 0. \quad (39)$$

Proof. For fixed i , its contribution is $\sum_{d=1}^{L_{i,X}(H)} (-1)^d$, equal to -1 for odd $L_{i,X}(H)$ and 0 for even $L_{i,X}(H)$. $\square$

Theorem 6.6 (Bounded-gap criterion). [E] *For fixed H , the following are equivalent:*

$$g_i \leq H \text{ for infinitely many } i, \quad - \sum_{h \leq H} C_X(h) \longrightarrow \infty.$$

Consequently, in the extended integers,

$$\inf \left\{ H : - \sum_{h \leq H} C_X(h) \rightarrow \infty \right\} = \liminf_{n \rightarrow \infty} g_n. \quad (40)$$

Proof. The divergence in (39) immediately implies infinitely many $g_i \leq H$. Conversely, if the right-hand side of (39) did not tend to infinity, then only finitely many of the limiting counts $L_i(H) := \#\{j > i : p_j - p_i \leq H\}$ could be odd: otherwise any prescribed finite set of odd limiting counts would eventually appear among the $L_{i,X}(H)$. Thus eventually every $L_i(H)$ is even. Whenever $g_i \leq H$, we must then have $L_i(H) \geq 2$, forcing $g_{i+1} \leq H$. Induction would make every sufficiently late gap at most H , implying $\pi(x) \gg x/H$, contrary to $\pi(x) = o(x)$. Taking the least integer threshold gives (40). $\square$

The cumulative identity can be sharpened at each individual lag. Let

$$\Pi_h(X) := \#\{p : p + h \leq X, p, p + h \in \mathbb{P}\},$$

and let $\Pi_{\{0,t,h\}}(X)$ count $p \leq X - h$ for which $p, p + t, p + h$ are all prime.

Theorem 6.7 (Prime-pair main term). [A+E] *For every fixed h ,*

$$0 \leq C_X(h) + \Pi_h(X) \leq 2 \sum_{1 \leq t < h} \Pi_{\{0,t,h\}}(X), \quad (41)$$

and hence

$$C_X(h) = -\Pi_h(X) + O_h\left(\frac{X}{\log^3 X}\right). \quad (42)$$

In particular,

$$C_X(2) = -\Pi_2(X) \quad (43)$$

exactly.

Proof. A pair at distance h contributes -1 when its prime-index difference is odd and $+1$ when that difference is even. After adding the unweighted count $\Pi_h(X)$, the first case contributes 0 and the second 2. An even index difference is at least 2, so there is an intermediate prime $p + t$, giving (41). A fixed-triple Selberg upper-bound sieve gives $\Pi_{\{0,t,h\}}(X) \ll_h X/\log^3 X$. For $h = 2$, the only internal offset is $t = 1$, and no three integers $p, p + 1, p + 2$ are all prime; hence the correction term vanishes. $\square$

Theorem 6.8 (Single-lag recovery of the least recurrent gap). [E] *Let*

$$H_* := \liminf_{n \rightarrow \infty} g_n.$$

For every fixed $h < 2H_$,*

$$C_X(h) = -\#\{n : p_{n+1} \leq X, g_n = h\} + O_h(1). \quad (44)$$

Therefore

$$H_* = \min\{h \geq 1 : C_X(h) \rightarrow -\infty\}. \quad (45)$$

Proof. Because the gaps are integral, all sufficiently late gaps are at least H_* , while $g_n = H_*$ occurs infinitely often. A sufficiently late prime pair at distance $h < 2H_*$ cannot contain an intermediate prime, since then its distance would be the sum of at least two gaps, each at least H_* . This proves (44). For $h < H_*$ there are only finitely many such pairs, so $C_X(h)$ is eventually constant; for $h = H_*$ the right side tends to $-\infty$. $\square$

The published Polymath8b bound $H_* \leq 246$ [20] now gives the stronger individual conclusion

$$\text{for some even } h \leq 246, \quad C_X(h) \longrightarrow -\infty. \quad (46)$$

This is full divergence, rather than merely a divergent liminf. It does not identify the lag h and does not imply the twin-prime conjecture $H_* = 2$.

7 Doubling orbits, Gauss orbits, and continuant ratios

7.1 Endpoint depth in the doubling orbit

At the k -th prime time define

$$x_k := \left\{ 2^{p_k-1}(2C_{PP}) \right\}, \quad d_k := k \pmod{2}, \quad \delta_k := |x_k - d_k|. \quad (47)$$

Thus $d_k \in \{0, 1\}$ and δ_k is the distance from x_k to the nearest endpoint of the unit interval.

Theorem 7.1 (Affine gap cocycle and binary decoder). [E] *For every $k \geq 1$,*

$$x_k = d_k(1 - 2^{-g_k}) + 2^{-g_k}x_{k+1}, \quad (48)$$

$$2^{-(g_k+1)} < \delta_k < 2^{-g_k}, \quad g_k = \lfloor -\log_2 \delta_k \rfloor. \quad (49)$$

Proof. The binary expansion of x_k begins with exactly g_k copies of d_k , followed by the opposite bit. Summing this initial run gives (48). If $d_k = 0$, then $x_{k+1} \in (1/2, 1)$; if $d_k = 1$, then $x_{k+1} \in (0, 1/2)$. The two strict bounds and the floor decoder follow. $\square$

7.2 The corrected Gauss-tail dictionary

Let

$$\alpha_P = [0; 2, g_1, g_2, \dots] \quad (50)$$

and write its convergent denominators as

$$q_{-1} = 0, \quad q_0 = 1, \quad q_1 = 2, \quad q_{n+1} = g_n q_n + q_{n-1} \quad (n \geq 1). \quad (51)$$

For the Gauss map $G(x) = \{1/x\}$, the correct indexing is

$$z_k := G^k(\alpha_P) = [0; g_k, g_{k+1}, \dots] \quad (k \geq 1). \quad (52)$$

Theorem 7.2 (Three exact integer parts). [E] *For every $k \geq 1$,*

$$\left\lfloor \frac{q_{k+1}}{q_k} \right\rfloor = \left\lfloor \frac{1}{z_k} \right\rfloor = \lfloor -\log_2 \delta_k \rfloor = g_k. \quad (53)$$

In particular,

$$\left| \frac{1}{z_k} + \log_2 \delta_k \right| < 1. \quad (54)$$

Proof. The recurrence gives $q_{k+1}/q_k = g_k + q_{k-1}/q_k$ with the second term in $(0, 1)$. The continued-fraction tail gives $1/z_k = g_k + z_{k+1}$ with $z_{k+1} \in (0, 1)$. The third equality is (49); the two real numbers in (54) lie in the same unit interval. $\square$

Theorem 7.3 (Bounded-gap dictionary). [E] *For a fixed integer $H \geq 1$, the following are equivalent:*

- (a) $-\sum_{h \leq H} C_X(h) \rightarrow \infty$;
- (b) $\delta_k > 2^{-(H+1)}$ for infinitely many k ;
- (c) $z_k > 1/(H+1)$ for infinitely many k ;
- (d) $\lfloor q_{k+1}/q_k \rfloor \leq H$ for infinitely many k .

Proof. By [theorem 6.6](#), (a) is equivalent to $g_k \leq H$ infinitely often. The equivalence with (b) is (49); that with (d) is (53). Finally $z_k = 1/(g_k + z_{k+1})$, with $0 < z_{k+1} < 1$, makes $g_k \leq H$ equivalent to $z_k > 1/(H + 1)$. $\square$

This theorem is a lossless coordinate translation of bounded gaps. It does not improve the numerical bounded-gap constant by itself.

7.3 Density-one cusp concentration and recurrent excursions

Lemma 7.4 (Small consecutive gaps). [\[A\]](#) *Uniformly for $2 \leq y \leq \log X$,*

$$\#\{k : p_{k+1} \leq X, g_k \leq y\} \ll \frac{Xy}{\log^2 X}. \quad (55)$$

Proof. Enlarge the count of consecutive pairs to all prime pairs at even distances $h \leq y$. The Selberg pair sieve gives

$$\#\{p \leq X : p, p+h \in \mathbb{P}\} \ll \frac{X}{\log^2 X} \prod_{\substack{\ell|h \\ \ell>2}} \frac{\ell-1}{\ell-2}.$$

Expanding the product as a squarefree divisor sum shows that its average over $h \leq y$ is $O(1)$, yielding (55). $\square$

For every fixed H ,

$$\#\{k \leq K : g_k \leq H\} = O_H(K/\log K). \quad (56)$$

Nevertheless the set is infinite for $H = 246$ by Polymath8b. Thus $g_k \rightarrow \infty$, $\delta_k \rightarrow 0$, $z_k \rightarrow 0$, and $g_{k+1}/g_k \rightarrow \infty$ each hold on a set of indices of natural density one, but none holds pointwise.

Let

$$\nu_K := \frac{1}{K} \sum_{k \leq K} \delta_{g_k}, \quad \eta_K := \frac{1}{K} \sum_{k \leq K} \delta_{z_k}.$$

Theorem 7.5 (Quantitative cusp concentration). [\[A\]](#) *For the usual metric on $[0, 1]$,*

$$W_1(\nu_K, \delta_0) \ll \frac{1}{\log K}, \quad W_1(\eta_K, \delta_0) \ll \frac{\log \log K}{\log K}. \quad (57)$$

Proof. Put $X = p_{K+1} \asymp K \log K$. From [lemma 7.4](#) and the prime number theorem,

$$\#\{k \leq K : g_k \leq y\} \ll K \frac{y}{\log K} \quad (2 \leq y \leq \log X).$$

Summation over dyadic gap layers gives

$$\sum_{k \leq K} 2^{-g_k} \ll \frac{K}{\log K}, \quad \sum_{k \leq K} \frac{1}{g_k} \ll \frac{K \log \log K}{\log K}.$$

Indeed the first series has a summable weight on layers, while each of the $O(\log \log K)$ dyadic layers up to $\log X$ contributes $O(K/\log K)$ to the second; the tail $g_k > \log X$ contributes $O(K/\log X)$. Now $\delta_k < 2^{-g_k}$ and $z_k < 1/g_k$. The W_1 distance from an empirical measure on $[0, 1]$ to δ_0 is the mean of its coordinates, proving the result. $\square$

7.4 A dimension bound for the Gauss-tail closure

Weak concentration at the cusp does not by itself control the closure of the exceptional tails. The residue graph modulo 6 supplies such a geometric bound.

Theorem 7.6 (Thin Gauss-tail closure). [\[A+E\]](#) For

$$Z := \overline{\{z_k : k \geq 2\}} \subset [0, 1],$$

one has

$$\dim_H Z \leq \frac{2}{3}. \quad (58)$$

Proof. For primes larger than 3, the residue of p_k modulo 6 is 1 or 5. The inverse branch associated with a gap g is $\phi_g(x) = 1/(g+x)$, with $|\phi'_g(x)| \leq g^{-2}$. At Hausdorff exponent $s = 2/3$, the two-state content matrix is dominated by

$$M = \begin{pmatrix} A & B \\ C & A \end{pmatrix},$$

where

$$A = \sum_{m \geq 1} (6m)^{-4/3}, \quad B = \sum_{m \geq 0} (6m+4)^{-4/3}, \quad C = \sum_{m \geq 0} (6m+2)^{-4/3}.$$

The integral test gives

$$A \leq \frac{4}{6^{4/3}}, \quad B \leq 4^{-4/3} + \frac{1}{2} 4^{-1/3}, \quad C \leq 2^{-4/3} + \frac{1}{2} 2^{-1/3}.$$

Consequently

$$\rho(M) = A + \sqrt{BC} < 0.980 < 1.$$

Iterating the graph-directed cylinder cover therefore makes its total $2/3$ -content tend to zero. The finitely many initial tails do not affect Hausdorff dimension, proving [\(58\)](#). $\square$

8 Prime-gap continuants and lacunary support series

8.1 OEIS continuants and logarithmic growth

Let

$$\beta := [1; g_1, g_2, \dots]$$

and let $A_n/B_n = [1; g_1, \dots, g_n]$ with offset $n = 0$. Then

$$\alpha_P = \frac{1}{1 + \beta}, \quad q_{n+1} = A_n + B_n. \quad (59)$$

Thus for $Q_m := q_{2m}$,

$$Q_m = A_{2m-1} + B_{2m-1}; \quad (60)$$

Q_m should not be identified with the denominator sequence B_n itself. The sequences A_n, B_n are OEIS A109139 and A109140 with their stated offset.

For positive partial quotients, the continuant recurrence gives the elementary sandwich

$$\prod_{j \leq n} a_j \leq K_n(a_1, \dots, a_n) \leq \prod_{j \leq n} (a_j + 1).$$

Here the exact recurrence gives a slightly sharper form.

Theorem 8.1 (Full-gap denominator growth). [A] As $K \rightarrow \infty$,

$$0 \leq \log q_{K+1} - \log 2 - \sum_{k \leq K} \log g_k \leq \sum_{k \leq K} \log \left(1 + \frac{1}{g_k}\right) \ll \frac{K \log \log K}{\log K}. \quad (61)$$

Consequently

$$\log q_n = n \log \log n + O(n), \quad \log Q_m = 2m \log \log m + O(m). \quad (62)$$

Proof. The continuant inequalities give the first two bounds in (61), and the layer estimate in the proof of theorem 7.5 bounds the last sum.

Let $X = p_{K+1} \asymp K \log K$. Integrating lemma 7.4 over the levels $g_k < \log X$ yields

$$\sum_{k \leq K} (\log \log X - \log g_k)_+ = O(K).$$

Hence $\sum_{k \leq K} \log g_k \geq K \log \log X - O(K)$. Jensen's inequality and $\sum_{k \leq K} g_k = p_{K+1} - 2$ give the reverse bound

$$\sum_{k \leq K} \log g_k \leq K \log \frac{p_{K+1} - 2}{K} = K \log \log K + O(K).$$

Combine these estimates with (61). □

8.2 Half gaps and irrationality exponents

Write the odd primes as $3 = r_1 < r_2 < \dots$ and set

$$h_n := \frac{r_{n+1} - r_n}{2}, \quad H_{-1} = 0, \quad H_0 = 1, \quad H_n = h_n H_{n-1} + H_{n-2}.$$

Theorem 8.2 (Half-gap growth and three irrationality exponents). [A]

$$\log H_n = n \log \log n + O(n), \quad (63)$$

and

$$\mu(\alpha_P) = \mu(\beta) = \mu([0; h_1, h_2, \dots]) = 2. \quad (64)$$

Proof. Replacing g_n by $h_n = g_{n+1}/2$ changes the relevant logarithmic sum by $O(n)$, so the proof of theorem 8.1 gives (63). For an infinite simple continued fraction,

$$\mu([a_0; a_1, a_2, \dots]) = 2 + \limsup_n \frac{\log a_{n+1}}{\log q_n}.$$

Now $\log g_n, \log h_n = O(\log n)$, whereas the denominator logarithms are $n \log \log n + O(n)$. The extra term is zero. The fractional-linear relation $\alpha_P = 1/(1 + \beta)$ preserves irrationality exponent. □

The exponent 2 does not decide algebraicity: algebraic irrationals and many transcendental numbers have the same irrationality exponent.

8.3 Congruence rigidity

Theorem 8.3 (Full-gap congruence and lacunarity). [E] *For every $m \geq 1$,*

$$Q_m \equiv 3 \pmod{4}, \quad Q_{m+1} > 5Q_m. \quad (65)$$

Proof. All g_k with $k \geq 2$ are even. Reducing (51) modulo 2 and 4 gives q_{2m-1} even and $q_{2m} \equiv q_{2m-2} \equiv 3 \pmod{4}$. Also $q_{2m+1} > 2q_{2m}$ and $q_{2m+2} > 2q_{2m+1} + q_{2m} > 5q_{2m}$. $\square$

Theorem 8.4 (Exclusion of the factor 3). [E] *For every $n \geq 1$,*

$$H_n \equiv -r_{n+1}H_{n-1} \pmod{3}, \quad 3 \nmid H_n. \quad (66)$$

Proof. The case $n = 1$ is direct. Since $2^{-1} \equiv -1 \pmod{3}$, $h_n \equiv r_n - r_{n+1} \pmod{3}$. For $n \geq 2$, substitute the induction hypothesis $H_{n-1} \equiv -r_n H_{n-2}$ and use $r_n^2 \equiv 1 \pmod{3}$. Iteration gives

$$H_n \equiv (-1)^n \prod_{j=2}^{n+1} r_j \pmod{3},$$

whose factors are all units modulo 3. $\square$

8.4 Natural boundaries and arithmetic values

Define

$$F_Q(z) := \sum_{m \geq 1} z^{Q_m}, \quad F_H(z) := \sum_{n \geq 0} z^{H_n} = 2z + \sum_{n \geq 2} z^{H_n}. \quad (67)$$

Theorem 8.5 (Support-series natural boundaries). [A] *Both F_Q and F_H have the unit circle as a natural boundary. They are transcendental over $\mathbb{C}(z)$ and are not D-finite.*

Proof. The inequality $Q_{m+1}/Q_m > 5$ permits the Hadamard gap theorem for F_Q . For the half-gap sequence,

$$H_{n+1} - H_n = (h_{n+1} - 1)H_n + H_{n-1} \geq H_{n-1} \rightarrow \infty.$$

Apart from the initial coefficient $[z]F_H(z) = 2$, its coefficients are 0 or 1. They cannot be eventually periodic because the support gaps tend to infinity while the support is infinite. The radius is 1, so Pólya–Carlson gives the natural boundary. The transcendence and non-D-finiteness consequences are as in theorem 3.10. $\square$

The congruence $Q_m \equiv 3 \pmod{4}$ also gives the exact rotation law

$$F_Q(iz) = -iF_Q(z), \quad F_Q(z) = z^3 G(z^4) \quad (68)$$

for a suitable power series G .

Theorem 8.6 (Transcendental algebraic specializations). [L] *For every algebraic number α with $0 < |\alpha| < 1$, both $F_Q(\alpha)$ and $F_H(\alpha)$ are transcendental.*

Proof. For F_Q , apply Corollary 5 of Corvaja–Zannier [9] to the Hadamard sequence Q_m . For F_H , use their Corollary 4: the exponent differences tend to infinity, while

$$\limsup_n \frac{H_{n+1}}{H_n} \geq \limsup_n h_{n+1} = \infty$$

because prime gaps are unbounded. The finite initial coefficient $2z$ is harmless. $\square$

Lemma 8.7 (Lacunary rational specializations). [E] *Let $e_1 < e_2 < \dots$ be integers such that*

$$e_{j+1} - e_j \rightarrow \infty, \quad \limsup_j \frac{e_{j+1}}{e_j} = \infty.$$

For every rational r with $0 < |r| < 1$, the number $\sum_j r^{e_j}$ is Liouville.

Proof. Write $r = a/b$ in lowest terms. The J -th partial sum has denominator dividing b^{e_J} , while

$$\left| \sum_{j>J} r^{e_j} \right| \leq \frac{|r|^{e_{J+1}}}{1 - |r|}$$

for large J ; when $r < 0$, the first omitted term dominates the remaining tail once the support gaps are large. Along a subsequence on which e_{J+1}/e_J is arbitrarily large, the error is smaller than every prescribed power of b^{-e_J} . The tail is nonzero, and the standard rational lower bound also rules out a rational limit. Thus there are infinitely many rational approximants of every exponent. $\square$

Corollary 8.8 (Liouville values of the two continuant series). [E] *For every rational r with $0 < |r| < 1$, both $F_Q(r)$ and $F_H(r)$ are Liouville numbers.*

Proof. For Q_m , the ratios exceed 5 and become arbitrarily large because $Q_{m+1}/Q_m > g_{2m}g_{2m+1}$. For H_n , $H_{n+1}/H_n > h_{n+1}$ has infinite limsup and the exponent differences tend to infinity. Apply lemma 8.7. $\square$

9 Gilbreath obstruction languages at fixed depth

This section concerns exact finite-state descriptions of obstruction languages, not a proof of Gilbreath's conjecture. Normalize the gaps by

$$c_j := \frac{p_{j+2} - p_{j+1}}{2} - 1, \quad (\mathcal{D}v)_j := |v_{j+1} - v_j|, \quad C_{r,j} := (\mathcal{D}^r c)_j. \quad (69)$$

The equivalence with the classical Gilbreath formulation and the relevance of long zero blocks and shallow $\{0, d\}$ blocks are due to Chase, Hunter and Tao [6]. We develop the inverse language and its exact finite-state compression.

9.1 Shallow arithmetic barriers

Proposition 9.1 (Depth-zero barriers). [E] *The only two consecutive zero entries of c are $c_1 = c_2 = 0$. More generally, if an odd prime ℓ divides d and $p_{j+1} > \ell$, then a $\{0, d\}$ block in c has at most $\ell - 2$ entries.*

Proof. Two consecutive zeros correspond to $p, p+2, p+4$, so reduction modulo 3 leaves only 3, 5, 7. In a $\{0, d\}$ block, every actual gap is $2(c_t + 1) \equiv 2 \pmod{\ell}$. If there were $\ell - 1$ gaps, their ℓ endpoints would visit every residue modulo ℓ , forcing one endpoint to be a multiple of ℓ larger than ℓ . $\square$

Proposition 9.2 (A depth-one primordial barrier). [A] *Suppose L consecutive entries of $\mathcal{D}c$ vanish, so that $L + 1$ consecutive entries of c have a common value $u \leq H$. If the first prime endpoint exceeds $L + 2$, then*

$$\prod_{\substack{3 \leq \ell \leq L+2 \\ \ell \text{ prime}}} \ell \mid u + 1. \quad (70)$$

Consequently $L = O(\log(H + 2))$.

Proof. The $L + 2$ prime endpoints form an arithmetic progression of common difference $2(u + 1)$. For each odd prime $\ell \leq L + 2$, a nonzero common difference modulo ℓ would make the endpoints hit residue 0. Hence every such ℓ divides $u + 1$. The prime number theorem for the Chebyshev function gives the logarithmic bound. $\square$

9.2 Affine inverse language and parity recurrence

Fix a bottom word $V^{(r)}$. A one-row inverse is specified by a starting value $C_s = V_0^{(s)}$ and signs $\varepsilon_t^{(s)} \in \{-1, 1\}$ through

$$V_{t+1}^{(s)} = V_t^{(s)} + \varepsilon_t^{(s)} V_t^{(s+1)}, \quad V_t^{(s)} \geq 0. \quad (71)$$

Theorem 9.3 (Affine polyhedral inverse decomposition). [E] *Fix r, K, d and a bottom word $V^{(r)} = d\eta$ with $\eta \in \{0, 1\}^K$. For each full sign pattern ε there are integer matrices $A_\varepsilon, B_\varepsilon$ such that*

$$V^{(0)} = A_\varepsilon C + dB_\varepsilon \eta. \quad (72)$$

The nonnegativity constraints are rational linear inequalities in (C, η) . Accordingly the integral top-row preimages form a finite union, over sign patterns, of affine images of lattice points in rational polyhedra.

Proof. Iterate (71) upward. For a fixed sign pattern every entry is affine in the row-start vector C and in $d\eta$; all remaining restrictions are precisely the nonnegativity inequalities. $\square$

Modulo 2 the signs disappear:

$$(\mathcal{D}^r c)_j \equiv \sum_{t=0}^r \binom{r}{t} c_{j+t} \pmod{2}. \quad (73)$$

If d is even and $\mathcal{D}^r c \in \{0, d\}$, then $(1 + E)^r c = 0$ over $\mathbb{F}_2$. With $P = 2^{\lceil \log_2 r \rceil}$, the Frobenius identity gives $c_{j+P} \equiv c_j \pmod{2}$ at all interior indices where the recurrence is defined. The order- r recurrence is determined by its first r bits, so there are at most 2^r parity seeds, independently of the block length.

9.3 Wheel coordinates and syndetic cusp escape

For a height bound H , put $B = H + 1$ and

$$u_0 = 0, \quad u_m = \sum_{t < m} (c_t + 1), \quad 1 \leq u_{m+1} - u_m \leq B.$$

Let S be a finite set of odd primes, $Q = \prod_{\ell \in S} \ell$, and let $\Delta(Q)$ be the largest cyclic gap between reduced residues modulo Q .

Theorem 9.4 (Exact wheel criterion). [E] *An infinite increasing wheel-survivor path with increments in $\{1, \dots, B\}$ exists if and only if*

$$\Delta(Q) \leq B. \quad (74)$$

Proof. For each $\ell \in S$ the primality constraint forbids one residue. The Chinese remainder theorem aligns these residues to a single translate $r + (\mathbb{Z}/Q\mathbb{Z})^\times$. If every cyclic gap is at most B , traverse successive allowed residues periodically. Conversely an infinite path must repeatedly cross every cyclic gap, so a gap larger than B is impossible. $\square$

Known lower bounds for long sifted intervals, in particular those of Ford–Green–Konyagin–Maynard–Tao [12], imply that for every sufficiently large B one may choose a finite wheel for which no infinite height- H survivor exists and whose finite-state path length is bounded by

$$L(B) \leq \exp\left(O\left(\frac{B \log_2 B}{\log B \log_3 B}\right)\right) = \exp(o(B)). \quad (75)$$

Here $\log_j$ denotes the j -fold iterated logarithm. The harmless factor 2 between integer gaps and the normalized half-gap coordinate has been absorbed into B . We do not use the stronger, previously uncited claim for a particular primorial wheel.

Corollary 9.5 (Syndetic cusp escape). [L+E] *For every fixed B there is $L(B) < \infty$ such that, sufficiently far out, every block of $L(B)$ consecutive prime gaps contains some $g_k > 2B$. At the same index,*

$$\delta_k < 2^{-2B}, \quad z_k < \frac{1}{2B}. \quad (76)$$

For large B , $L(B)$ may be chosen as in (75).

Proof. A block with all $g_k \leq 2B$ gives a normalized survivor with increments at most B . After the finitely many wheel primes, actual prime endpoints avoid all forbidden residues. The acyclic finite wheel therefore bounds the path length. Apply (49) and $z_k < 1/g_k$. $\square$

9.4 Anchored congruence states and transfer matrices

Fix r, H and a finite wheel S . Let the current right difference boundary be $E = (e_0, \dots, e_{r-1})$. Appending $a \in \{0, \dots, H\}$ gives

$$f_0 = a, \quad f_k = |f_{k-1} - e_{k-1}| \quad (1 \leq k \leq r). \quad (77)$$

The new boundary is $(f_0, \dots, f_{r-1})$ and the new bottom label is f_r . Fix now an actual starting prime P . For every $\ell \in S$, let

$$x_\ell := P + \text{current actual offset} \pmod{\ell}. \quad (78)$$

Appending a gives

$$x'_\ell = x_\ell + 2(a+1) \pmod{\ell}, \quad x'_\ell \neq 0. \quad (79)$$

Theorem 9.6 (Anchored label-preserving right congruence). [E] *For fixed (r, H, S, P) , the history map*

$$\text{history} \mapsto (E, (x_\ell)_{\ell \in S})$$

has a label-preserving right-congruence kernel on the set of already wheel-admissible histories. Equal states have, for every append value, the same bottom label, the same wheel admissibility, the same next state, and hence identical admissible labelled futures.

Proof. The bottom label and the next difference boundary are determined by (77); the anchored congruence decision and next residues are determined by (79). Induction on continuation length proves the assertion. $\square$

Remark 9.7 (Why the endpoint residue is necessary). The relative-mask state used in Version 2.0 is not a right congruence for a fixed actual start. Take $r = 1$, $H = 2$, $S = \{5\}$, and $P \equiv 1 \pmod{5}$. The histories

$$h_1 = (0, 2, 1, 2), \quad h_2 = (2, 0, 0, 2)$$

have actual-offset trajectories modulo 5

$$(0, 2, 3, 2, 3), \quad (0, 1, 3, 0, 1),$$

respectively. Both have $E = (2)$ and the same translated visited mask $C_5 = \{0, 2, 4\}$. Appending $a = 2$ gives the same bottom label 0, but the first endpoint is $1 + 4 \equiv 0 \pmod{5}$ and is forbidden, whereas the second is $1 + 2 \equiv 3 \pmod{5}$ and is allowed. Relative masks remain appropriate for the different, existential question whether some starting translate avoids all visited residues; they cannot replace (78) in the anchored language.

The smallest wheel already gives a uniform entropy saving. Let $n_j = \#\{0 \leq a \leq H : a \equiv j \pmod{3}\}$. On the two nonzero endpoint residues modulo 3, the anchored transition matrix is

$$A_{3,H} = \begin{pmatrix} n_2 & n_1 \\ n_0 & n_2 \end{pmatrix}, \quad \rho(A_{3,H}) = n_2 + \sqrt{n_0 n_1} \leq \frac{2(H+1)}{3}. \quad (80)$$

Thus the number of length- K height words surviving even this single congruence is at most $\exp[-(\log(3/2) + o(1))K]$ times the full $(H+1)^K$ word count. This is a uniform congruence saving, not a proof that every surviving word is realized by primes.

For $d \geq 1$ define

$$N_{r,d}(K, H) := \#\{c \in \{0, \dots, H\}^{K+r} : \mathcal{D}^r c \in \{0, d\}^K\}. \quad (81)$$

Theorem 9.8 (Finite transfer representation). [E] *There exist a finite nonnegative matrix $Q_{r,H,d}$ and an initial multiplicity vector $\mathbf{m}$ such that*

$$N_{r,d}(K, H) = \mathbf{m}^T Q_{r,H,d}^K \mathbf{1}. \quad (82)$$

Moreover

$$N_{r,d}(K, H) \leq (H+1)^r (3 \cdot 2^{r-1})^K. \quad (83)$$

Proof. The right boundary in (77) has finitely many possibilities and append is a Markov transition. In reverse, a prescribed bottom label 0 has at most 2^{r-1} sign choices and a label d at most 2^r , giving the per-step branching bound $3 \cdot 2^{r-1}$. $\square$

Define the fixed-height entropy by

$$h_{r,d}(H) := \limsup_{K \rightarrow \infty} \frac{1}{K} \log N_{r,d}(K, H).$$

For $r = 1$, direct path-graph diagonalization gives

$$\rho(Q_{1,H,d}) = 1 + 2 \cos \frac{\pi}{\lfloor H/d \rfloor + 2} < 3.$$

For the next depth one obtains the explicit domination

$$\rho(Q_{2,H,d}) \leq 2 + 4 \cos \frac{\pi}{\lfloor H/d \rfloor + 2} < 6, \quad (84)$$

and hence

$$h_{2,d}(H) \leq \log \left(2 + 4 \cos \frac{\pi}{\lfloor H/d \rfloor + 2} \right).$$

Indeed the dominating gap-state matrix has entries

$$W_{y,t} = m(t) \mathbf{1}_{|t-y| \in \{0,d\}}, \quad m(0) = 1, \quad m(t) = 2 \ (t > 0),$$

and each residue component is a finite path with a self-loop. Its adjacency eigenvalue is

$$2 \cos \left(\frac{\pi}{\lfloor H/d \rfloor + 2} \right),$$

and the multiplicity weight produces (84).

These finite-state results count and compress potential obstruction words. They do not prove that every actual prime input avoids them, and therefore do not settle Gilbreath's conjecture.

10 The dyadic numerator cocycle

10.1 Exact recurrence

Let

$$C_n := \sum_{k=1}^n (-1)^{k+1} 2^{-p_k} = \frac{N_n}{2^{p_n}}. \quad (85)$$

The numerator sequence (N_n) is recorded as OEIS A122150.

Theorem 10.1 (Dyadic numerator recurrence). [E]

$$N_1 = 1, \quad N_n = 2^{g_{n-1}} N_{n-1} + (-1)^{n+1} \quad (n \geq 2). \quad (86)$$

Every N_n is positive and odd, so 2^{p_n} is the exact reduced denominator in (85).

Proof. Raise C_{n-1} to denominator 2^{p_n} and add the final term. The first summand in (86) is even and the second odd. Positivity also follows from the alternating series with strictly decreasing term magnitudes. $\square$

10.2 Autonomous and indexed decoding; non-holonomicity

For odd $x > 1$, put

$$\chi_4(x) := (-1)^{(x-1)/2}, \quad \mathcal{D}_N(x) := \text{oddpart}(x - \chi_4(x)),$$

where $\text{oddpart}(m) = m/2^{v_2(m)}$.

Theorem 10.2 (Autonomous numerator decoder). [E] For every $n \geq 3$,

$$\mathcal{D}_N(N_n) = N_{n-1}, \quad g_{n-1} = v_2(N_n^2 - 1) - 1. \quad (87)$$

Moreover,

$$g_{n-1} = \lfloor \log_2 N_n \rfloor - \lfloor \log_2 N_{n-1} \rfloor. \quad (88)$$

Thus a single integer N_n , together with the fixed seed $(p_1, p_2) = (2, 3)$, determines n , all preceding gaps, and all primes $p_1, \dots, p_n$.

Proof. Write $g = g_{n-1}$, $M = N_{n-1}$, and $\varepsilon = (-1)^{n+1}$. Since $g \geq 2$ for $n \geq 3$, the recurrence $N_n = 2^g M + \varepsilon$ gives $N_n \equiv \varepsilon \pmod{4}$, hence $\chi_4(N_n) = \varepsilon$. This proves the first identity. Also

$$N_n^2 - 1 = 2^{g+1} M (2^{g-1} M + \varepsilon),$$

whose final factor is odd, proving the valuation formula.

If $2^d \leq M < 2^{d+1}$, then $2^{d+g} \leq 2^g M + \varepsilon < 2^{d+g+1}$. In the only marginal case $\varepsilon = -1$, the odd number $M > 1$ is strictly larger than 2^d . This proves (88). Iterating $\mathcal{D}_N$ terminates at $N_2 = 1$; the number of iterations determines n , while the displayed valuation recovers the gaps and $g_1 = 1$ is fixed. $\square$

Let $\epsilon_n = n \bmod 2$, and let a^m denote a run of m copies of the binary digit a . The stronger finite-word form is

$$(N_n - \epsilon_n)_2 = 1^{g_1} 0^{g_2} 1^{g_3} 0^{g_4} \dots ((n-1) \bmod 2)^{g_{n-1}} \quad (n \geq 2). \quad (89)$$

Indeed, an odd n appends g_{n-1} zeroes in the recurrence, while an even n uses $N_n = 2^{g_{n-1}}(N_{n-1} - 1) + (2^{g_{n-1}} - 1)$ and appends that many ones.

Theorem 10.3 (Non-holonomicity of OEIS A122150). [A+E] *The sequence (N_n) is not P -recursive. Equivalently, its ordinary generating series*

$$\mathcal{N}(t) := \sum_{n \geq 1} N_n t^n$$

is not D -finite as a formal power series.

Proof. The recurrence gives

$$\frac{N_{n+1}}{N_n} \geq 2^{g_n} - 1.$$

The large-gap theorem of Ford–Green–Konyagin–Maynard–Tao [12] implies $\limsup g_n / \log n = \infty$. Hence this ratio exceeds every fixed power of n along a subsequence.

If the positive increasing sequence (N_n) satisfied $\sum_{j=0}^d P_j(n) N_{n+j} = 0$ with $P_d \not\equiv 0$, then for all large n

$$\frac{N_{n+d}}{N_{n+d-1}} \leq \sum_{j < d} \left| \frac{P_j(n)}{P_d(n)} \right| \frac{N_{n+j}}{N_{n+d-1}} \leq n^{O(1)},$$

because the sequence is increasing and rational functions have polynomial growth. This contradicts the subsequence above. The equivalence with formal D -finiteness is standard [24]. $\square$

Theorem 10.4 (2-adic gap decoder). [E] *For every $n \geq 2$,*

$$g_{n-1} = v_2(N_n - (-1)^{n+1}). \quad (90)$$

Thus the numerator sequence alone recovers every prime:

$$p_n = 2 + \sum_{j=2}^n v_2(N_j - (-1)^{j+1}). \quad (91)$$

Proof. Subtract the final sign in (86). Since N_{n-1} is odd, the exact 2-adic valuation is g_{n-1} . Sum the recovered gaps. $\square$

Corollary 10.5 (A modulo-24 twin-gap sensor). [E] For $n \geq 3$,

$$g_{n-1} = 2 \iff \begin{cases} N_n \equiv 5 \pmod{24}, & n \text{ odd}, \\ N_n \equiv 19 \pmod{24}, & n \text{ even}. \end{cases} \quad (92)$$

Proof. For $n \geq 3$, g_{n-1} is even. Modulo 3, $2^{g_{n-1}} \equiv 1$, and (86) gives $N_n \equiv 2$ for odd n and $N_n \equiv 1$ for even n . Modulo 8, $2^g \equiv 4$ when $g = 2$ and 0 when $g \geq 4$. Combine the two conditions by the Chinese remainder theorem. $\square$

The sensor is exact but, like theorem 6.4, does not provide a lower bound for the number of twin gaps.

10.3 Gap-polynomial resonance and fixed-lag separation

For $1 \leq m < n$ put

$$R_{m,n} := \sum_{k=m+1}^n (-1)^{k+1} 2^{p_n - p_k}. \quad (93)$$

Its final summand is odd and all preceding summands are even, so $R_{m,n}$ is odd and nonzero.

Theorem 10.6 (Gap-polynomial resonance). [E]

$$N_n = 2^{p_n - p_m} N_m + R_{m,n}, \quad \gcd(N_n, N_m) = \gcd(N_m, R_{m,n}). \quad (94)$$

If $n = m + 2$ and an odd prime ℓ divides N_m , then

$$\ell \mid N_{m+2} \iff \text{ord}_\ell(2) \mid g_{m+1}. \quad (95)$$

Proof. Separate the first m terms of (85) after raising them to denominator 2^{p_n} , and apply the Euclidean algorithm. For $n = m + 2$, $R_{m,m+2} = \pm(2^{g_{m+1}} - 1)$, proving (95). $\square$

Corollary 10.7 (Fixed-lag logarithmic coprimality). [A+E] For every fixed $r \geq 1$,

$$\gcd(N_n, N_{n-r}) \leq |R_{n-r,n}| \leq r 2^{p_n - p_{n-r+1}} = N_n^{o(1)}. \quad (96)$$

Proof. The first two inequalities follow from (94) and (93). The prime number theorem implies $p_n - p_{n-r+1} = o(p_n)$ for fixed r . Moreover $N_n/2^{p_n} = C_n \rightarrow C_{PP} > 0$, so $\log N_n \sim p_n \log 2$. $\square$

10.4 The third support series

Because $N_1 = N_2 = 1$, write

$$F_N(z) := \sum_{n \geq 1} z^{N_n} = 2z + \sum_{n \geq 3} z^{N_n}. \quad (97)$$

Theorem 10.8 (Dyadic-numerator support). [A+L] The unit circle is a natural boundary for F_N ; in particular F_N is transcendental over $\mathbb{C}(z)$ and not D -finite. For every algebraic α with $0 < |\alpha| < 1$, the value $F_N(\alpha)$ is transcendental. For every rational r with $0 < |r| < 1$, the value $F_N(r)$ is a Liouville number.

Proof. For $n \geq 4$,

$$\frac{N_n}{N_{n-1}} = 2^{g_{n-1}} + \frac{(-1)^{n+1}}{N_{n-1}} \geq 4 - \frac{1}{5} > 1.$$

After the duplicate initial exponent is removed, the Hadamard gap theorem gives the natural boundary. Corollary 5 of Corvaja–Zannier [9] gives transcendence at nonzero algebraic points of the disk. Finally the ratios have infinite limsup because the prime gaps are unbounded, and the exponent differences tend to infinity. Apply lemma 8.7. $\square$

Corollary 10.9 (Arithmetic Picard fibres of F_N). [L+E] *Let $a \in \overline{\mathbb{Q}}$, let $\zeta \in \partial\mathbb{D}$, and let U be any neighborhood of ζ . Then*

$$\{z \in U \cap \mathbb{D} : F_N(z) = a\}$$

contains infinitely many transcendental numbers. In particular, every point of the unit circle is an accumulation point of transcendental zeroes of F_N .

Proof. After combining the duplicated initial exponent, F_N is an unbounded Hadamard-lacunary series. Murai’s sectorial Picard theorem [39, Remark 21] says that it assumes every complex value infinitely often in every boundary sector. The preimage of a value is discrete, so the infinitely many solutions in a fixed sector leave every compact subset of $\mathbb{D}$. Choosing first an angular sector and then an outer annulus contained in U proves the claim. Any nonzero algebraic solution would contradict theorem 10.8; when $a = 0$, the algebraic solution $z = 0$ accounts for at most one point. $\square$

Combining theorems 8.6 and 10.8 and corollary 8.8 gives the uniform value statement

$$\frac{F_Q, F_H, F_N}{\left| \begin{array}{c|c} 0 < |\alpha| < 1, \alpha \in \overline{\mathbb{Q}} & 0 < |r| < 1, r \in \mathbb{Q} \\ \hline \text{transcendental} & \text{Liouville.} \end{array} \right|} \quad (98)$$

These are values of support series, not the continued-fraction numbers in (64); the two irrationality-exponent statements concern different numbers.

Theorem 10.10 (Hypertranscendence of the three support series). [L+E] *Each of F_Q, F_H, F_N is hypertranscendental over $\mathbb{C}(z)$: none satisfies a nonzero algebraic differential equation*

$$P(z, F, F', \dots, F^{(d)}) = 0, \quad P \in \mathbb{C}[z, X_0, \dots, X_d].$$

Proof. The gap theorem of Lipshitz and Rubel [37, Theorem 4.1] excludes differential algebraicity for a nonpolynomial Hadamard-gap series. It applies directly to F_Q , because $Q_{m+1} > 5Q_m$, and to F_N , whose successive distinct exponents have ratio greater than 3 from some point onward.

For F_H , let $\omega^3 = 1$ and use the root-of-unity filters

$$F_{H,r}(z) := \frac{1}{3} \sum_{j=0}^2 \omega^{-rj} F_H(\omega^j z), \quad r = 1, 2.$$

By (66), $F_H = F_{H,1} + F_{H,2}$. Within either infinite filtered support, successive exponents have ratio greater than 2: if their original indices are adjacent and the residues agree, the corresponding h cannot equal 1, while skipping an index gives $H_{n+2} > 2H_n$. At least one filter is infinite, so it is a Hadamard-gap series. Differentially algebraic functions are closed under root-of-unity substitution and finite sums. If F_H were differentially algebraic, both filters would be, contradicting the gap theorem for the infinite one. $\square$

Finally, (85) proves the OEIS limit

$$\frac{N_n}{2^{p_n}} \longrightarrow C_{PP}.$$

Together with (6), it also exposes the one-bit shift in the binary expansion comment: A071986 is the digit word of $2C_{PP}$, whereas C_{PP} has one additional leading zero after the radix point.

11 The Mellin boundary of the dyadic numerator series

Set $C = C_{PP}$. By (85),

$$N_n = C_n 2^{p_n}, \quad C_n \longrightarrow C > 0,$$

and the alternating-series remainder gives

$$|C_n - C| \leq 2^{-p_{n+1}}. \quad (99)$$

Attach to the numerator cocycle the Dirichlet series

$$Z_N(s) := \sum_{n \geq 1} N_n^{-s}, \quad \Re s > 0, \quad (100)$$

where positive real logarithms are used. Since $\log N_n \sim p_n \log 2$, its abscissa of absolute convergence is 0.

11.1 Mellin representation and prime-support decomposition

Let

$$P_{\text{pr}}(z) := \sum_{p \in \mathbb{P}} z^p, \quad |z| < 1. \quad (101)$$

Theorem 11.1 (Mellin representation). [E] For $\Re s > 0$,

$$\Gamma(s) Z_N(s) = \int_0^\infty t^{s-1} F_N(e^{-t}) dt. \quad (102)$$

Proof. For $\sigma = \Re s > 0$, Tonelli's theorem applies because

$$\int_0^\infty t^{\sigma-1} \sum_{n \geq 1} e^{-t N_n} dt = \Gamma(\sigma) \sum_{n \geq 1} N_n^{-\sigma} < \infty.$$

Termwise Mellin integration proves (102); the multiplicity-two initial exponent in F_N agrees with $N_1 = N_2 = 1$. $\square$

Proposition 11.2 (Prime-support decomposition). [E] There is a function $\mathcal{R}$, holomorphic on $\Re s > -1$, such that

$$Z_N(s) = C^{-s} P_{\text{pr}}(2^{-s}) + \mathcal{R}(s) \quad (\Re s > 0), \quad (103)$$

where

$$\mathcal{R}(s) = \sum_{n \geq 1} 2^{-s p_n} (C_n^{-s} - C^{-s}). \quad (104)$$

The latter series converges normally on compact subsets of $\Re s > -1$.

Proof. Let K be compact in $\Re s > -1$ and put $\sigma_0 = \inf_{s \in K} \Re s > -1$. The derivatives of $x \mapsto x^{-s}$ are uniformly bounded for $s \in K$ on an interval containing C and all late C_n . Thus (99) gives

$$|2^{-sp_n}(C_n^{-s} - C^{-s})| \ll_K 2^{-\sigma_0 p_n - p_{n+1}} \leq 2^{-(1+\sigma_0)p_n}.$$

The majorant is summable. Normal convergence proves the assertion. $\square$

Theorem 11.3 (Natural boundary of Z_N). [A+E] *The imaginary axis $\Re s = 0$ is a natural boundary for Z_N : there is no holomorphic continuation across any point of that line. Consequently, Z_N has no meromorphic continuation through a nonempty open subarc of it.*

Proof. The integer-coefficient series P_{pr} has radius one and is not rational. Indeed, its bounded coefficient sequence cannot be eventually periodic: if it had eventual period q , then a sufficiently large prime p would force the composite number $p + pq = p(1 + q)$ to be prime. Pólya–Carlson [21, 5] makes the unit circle a natural boundary for P_{pr} .

If Z_N continued through a point of $\Re s = 0$, then $P_{\text{pr}}(2^{-s}) = C^s(Z_N(s) - \mathcal{R}(s))$ would continue there. The map $s \mapsto 2^{-s}$ has nonzero derivative and a local inverse, contradicting the natural boundary of P_{pr} . A meromorphic continuation through an open arc would be holomorphic near a non-pole point of that arc and is therefore excluded as well. $\square$

11.2 The inverse-logarithmic Gumbel fingerprint

For $m \geq 0$, define

$$A_m := (-1)^m \Gamma^{(m)}(1) = \int_0^\infty e^{-u} (-\log u)^m du. \quad (105)$$

The first values are

$$A_0 = 1, \quad A_1 = \gamma, \quad A_2 = \gamma^2 + \frac{\pi^2}{6}, \quad A_3 = \gamma^3 + \frac{\gamma\pi^2}{2} + 2\zeta(3). \quad (106)$$

Theorem 11.4 (All-orders boundary expansion). [A+E] *Let $y = s \log 2$ and $\Lambda = \log(1/y)$, with $s > 0$. For every fixed $M \geq 0$,*

$$yZ_N(s) = \sum_{m=0}^M \frac{A_m}{\Lambda^{m+1}} + O_M(\Lambda^{-M-2}) \quad (s \downarrow 0). \quad (107)$$

Equivalently, in the Poincaré sense,

$$(s \log 2)Z_N(s) \sim \sum_{m \geq 0} \frac{(-1)^m \Gamma^{(m)}(1)}{(\log(1/(s \log 2)))^{m+1}}. \quad (108)$$

Proof. Write $S(y) = P_{\text{pr}}(e^{-y}) = \sum_p e^{-yp}$. The classical zero-free-region form of the prime number theorem gives

$$\pi(x) = \text{Li}(x) + O(xe^{-c\sqrt{\log x}}). \quad (109)$$

Stieltjes integration and one integration by parts give, for some $c_1 > 0$,

$$yS(y) = \int_{2y}^\infty \frac{e^{-u}}{\Lambda + \log u} du + O(e^{-c_1\sqrt{\Lambda}}). \quad (110)$$

Split the integral at $e^{-\Lambda/2}$ and $e^{\Lambda/2}$. The exterior pieces are smaller than every negative power of Λ . In the middle range the finite geometric expansion is

$$\frac{1}{\Lambda + \log u} = \sum_{m=0}^M \frac{(-\log u)^m}{\Lambda^{m+1}} + O_M \left(\frac{|\log u|^{M+1}}{\Lambda^{M+2}} \right).$$

Every logarithmic moment against $e^{-u} du$ is finite, so

$$yP_{\text{pr}}(e^{-y}) = \sum_{m=0}^M \frac{A_m}{\Lambda^{m+1}} + O_M(\Lambda^{-M-2}). \quad (111)$$

Finally, $C^{-s} = 1 + O(s)$ and $\mathcal{R}$ is bounded near zero. The resulting corrections to $yZ_N(s)$ are smaller than every negative power of Λ , proving (107). $\square$

Corollary 11.5 (Exact boundary-moment recovery). [E] *For every $m \geq 0$,*

$$A_m = \lim_{s \downarrow 0} \Lambda^{m+1} \left((s \log 2) Z_N(s) - \sum_{j=0}^{m-1} \frac{A_j}{\Lambda^{j+1}} \right), \quad \Lambda = \log \frac{1}{s \log 2}. \quad (112)$$

In particular,

$$\gamma = \lim_{s \downarrow 0} \Lambda^2 \left((s \log 2) Z_N(s) - \frac{1}{\Lambda} \right), \quad (113)$$

and theorem 11.1 gives the equivalent F_N -formula

$$\gamma = \lim_{s \downarrow 0} \Lambda^2 \left[\frac{s \log 2}{\Gamma(s)} \int_0^\infty t^{s-1} F_N(e^{-t}) dt - \frac{1}{\Lambda} \right]. \quad (114)$$

Proof. Subtract the first m terms in theorem 11.4 and multiply the remainder by Λ^{m+1} . $\square$

Remark 11.6 (Gumbel interpretation). If U is exponential with mean one and $G = -\log U$, then G has the standard Gumbel distribution and

$$A_m = \mathbb{E}(G^m), \quad \mathbb{E}(e^{tG}) = \Gamma(1-t) \quad (t < 1).$$

Its cumulants are

$$\kappa_1 = \gamma, \quad \kappa_m = (m-1)! \zeta(m) \quad (m \geq 2), \quad (115)$$

because

$$\log \Gamma(1-t) = \gamma t + \sum_{m \geq 2} \frac{\zeta(m)}{m} t^m. \quad (116)$$

Thus $\zeta(2) = A_2 - A_1^2$ and $2\zeta(3) = A_3 - 3A_2A_1 + 2A_1^3$. These are boundary-coefficient identities; they do not decide the irrationality of γ or the transcendence of an individual odd zeta value.

11.3 Boundary moments and a regularized Gamma determinant

Let $\mathcal{H} = \ell^2(\mathbb{N})^{\oplus 5}$ and define

$$A_\Gamma := \text{diag} \left(\frac{1}{n}, \frac{1}{n}, \frac{1}{n}, -\frac{1}{n}, -\frac{1}{n} : n \geq 1 \right). \quad (117)$$

This compact self-adjoint operator is Hilbert–Schmidt but not trace class.

Theorem 11.7 (Boundary–Gamma determinant correspondence). [E] *For the truncation to $n \leq N$,*

$$\text{FPTr } A_\Gamma := \lim_{N \rightarrow \infty} (\text{Tr } A_{\Gamma, N} - \log N) = A_1 = \gamma, \quad (118)$$

whereas, for $m \geq 2$,

$$\text{Tr } A_\Gamma^m = (3 + 2(-1)^m)\zeta(m) = \frac{3 + 2(-1)^m}{(m-1)!} \kappa_m. \quad (119)$$

Moreover,

$$\det_2(I - tA_\Gamma) = \frac{e^{\gamma t}}{\Gamma(1-t)^3 \Gamma(1+t)^2}. \quad (120)$$

If $\mathcal{B}(t) = \sum_{m \geq 0} A_m t^m / m! = \Gamma(1-t)$, then

$$\mathcal{B}(t)^3 \mathcal{B}(-t)^2 = e^{A_1 t} \det_2(I - tA_\Gamma)^{-1}. \quad (121)$$

Thus the inverse-logarithmic fingerprint of Z_N and this regularized trace tower determine one another.

Proof. $\text{Tr } A_{\Gamma, N} = (3-2)H_N$, which proves (118); absolute summation proves (119). The two Weierstrass products

$$\prod_{n \geq 1} (1 - t/n) e^{t/n} = \frac{e^{\gamma t}}{\Gamma(1-t)}, \quad \prod_{n \geq 1} (1 + t/n) e^{-t/n} = \frac{e^{-\gamma t}}{\Gamma(1+t)}$$

give (120); see [42]. The moment–cumulant transformations are mutually inverse universal polynomial maps, which proves the final assertion. $\square$

11.4 Mahler exclusion and algebraic separation

Theorem 11.8 (All-base non-Mahler property). [L+E] *For no integer $k \geq 2$ is $F_N(z)$ a k -Mahler function.*

Proof. Write $F_N(z) = \sum_{m \geq 0} a_m z^m$. If its finite-valued coefficient sequence were k -automatic, the canonical base- k representations of the support would form an infinite regular language. The pumping lemma forces such a language to have at least cL words of length at most L . Here, however,

$$\#\{n : N_n < k^L\} \sim \frac{\log k}{\log 2} \frac{L}{\log L} = o(L), \quad (122)$$

because $\log N_n \sim p_n \log 2$. Thus (a_m) is not k -automatic. A finite-valued k -Mahler series in characteristic zero is k -automatic by Adamczewski–Bell–Smertnig [27, Theorem 11.1], giving the contradiction. $\square$

For a rational r with $0 < |r| < 1$, put $L_r := F_N(r)$.

Theorem 11.9 (Liouville– π algebraic independence). [L+E] *For every such r ,*

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(L_r, \pi) = 2. \quad (123)$$

Consequently L_r is algebraically independent from every $\zeta(2m)$, and

$$\overline{\mathbb{Q}}(L_r, \zeta(2), \zeta(4), \dots) = \overline{\mathbb{Q}}(L_r, \pi^2)$$

has transcendence degree 2.

Proof. By [theorem 10.8](#), L_r is a Liouville number and hence a Mahler U -number. Mahler's classification theorem says that algebraically dependent transcendental numbers have the same class, while π is an S - or T -number; see [\[31, Theorems 2.3 and 2.5\]](#). This proves [\(123\)](#). Euler's formula expressing $\zeta(2m)$ as a nonzero rational multiple of π^{2m} gives the rest. $\square$

For example, the explicit number

$$\frac{6}{\pi^2} \left(1 + 2^{-5} + 2^{-19} + 2^{-305} + 2^{-1219} + \dots \right) = \frac{6F_N(1/2)}{\pi^2} \quad (124)$$

is transcendental.

Theorem 11.10 (Relative algebraic independence of boundary jets). [\[L+E\]](#) Fix $r \in \mathbb{Q}$, $0 < |r| < 1$, and put

$$\tau_r(N) := \# \left\{ 1 \leq m \leq N : \text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(L_r, A_m) = 2 \right\}.$$

Then

$$\tau_r(N) \geq \left\lceil \sqrt{2 \lfloor N/2 \rfloor + \frac{9}{4}} - \frac{3}{2} \right\rceil. \quad (125)$$

In particular, infinitely many Gumbel boundary coefficients are algebraically independent from L_r .

Proof. Put $K = \overline{\mathbb{Q}}(L_r)$ and $M = \lfloor N/2 \rfloor$. The reflection formula $\Gamma(1-t)\Gamma(1+t) = \pi t / \sin \pi t$ gives, for $0 \leq k \leq M$,

$$\sum_{j=0}^{2k} \frac{(-1)^j A_j A_{2k-j}}{j!(2k-j)!} = c_k \pi^{2k}, \quad c_k \in \mathbb{Q}^\times. \quad (126)$$

Let I consist of those indices $1 \leq j \leq N$ for which A_j is transcendental over K , and write $t = |I|$. Adjoining the remaining A_j makes only a finite algebraic extension E/K . By [theorem 11.9](#), the numbers $1, \pi^2, \dots, \pi^{2M}$ are linearly independent over E . Equations [\(126\)](#) place them in the E -space of polynomials of total degree at most two in the t elements indexed by I , whose dimension is at most $\binom{t+2}{2}$. Hence $M+1 \leq (t+1)(t+2)/2$, which rearranges to [\(125\)](#). This is the relative version of the standard reflection-and-dimension argument for Gamma derivatives; compare [\[40\]](#). $\square$

Corollary 11.11 (The first four boundary coefficients). [\[E\]](#) At least two of A_1, A_2, A_3, A_4 are algebraically independent from L_r . In particular, at least one of A_2, A_3, A_4 is algebraically independent from L_r .

Proof. Besides $A_2 - A_1^2 = \zeta(2)$, the first four moments satisfy

$$5A_4 - 20A_1A_3 - 27A_2^2 + 84A_1^2A_2 - 42A_1^4 = 0.$$

This follows by substituting the complete Bell-polynomial expressions for the Gamma derivatives and using $\zeta(4) = \frac{2}{5}\zeta(2)^2$. If exactly one of A_1, A_2, A_3, A_4 were transcendental over $\overline{\mathbb{Q}}(L_r)$, the displayed relation is a nonconstant polynomial in that element (for A_3 , use $A_1 = \gamma \neq 0$), making it algebraic as well; if none were transcendental, all four would already be algebraic. Then $A_2 - A_1^2 = \pi^2/6$ would make π algebraic over $\overline{\mathbb{Q}}(L_r)$, contradicting [theorem 11.9](#). $\square$

12 The alternating prime-index Dirichlet series

Define

$$\mathcal{P}_{\text{alt}}(s) := \sum_{k \geq 1} (-1)^{k+1} p_k^{-s} = \sum_{j \geq 1} (p_{2j-1}^{-s} - p_{2j}^{-s}), \quad (127)$$

and let $\xi(x)$ be the indicator of $\bigcup_{j \geq 1} [p_{2j-1}, p_{2j})$.

12.1 Laplace–Mellin representation

Theorem 12.1 (Analyticity and complete monotonicity). [E] *The function $\mathcal{P}_{\text{alt}}$ is holomorphic on $\Re s > 0$ and*

$$\mathcal{P}_{\text{alt}}(s) = s \int_2^\infty \xi(x) x^{-s-1} dx. \quad (128)$$

For real $\sigma > 0$ it is positive, and $F(\sigma) = \mathcal{P}_{\text{alt}}(\sigma)/\sigma$ is strictly completely monotone.

Proof. For each pair,

$$p_{2j-1}^{-s} - p_{2j}^{-s} = s \int_{p_{2j-1}}^{p_{2j}} x^{-s-1} dx.$$

The disjoint intervals make the sum locally uniformly dominated on every half-plane $\Re s \geq \delta > 0$, proving (128). With $x = e^t$,

$$(-1)^r F^{(r)}(\sigma) = \int_{\log 2}^\infty \xi(e^t) t^r e^{-\sigma t} dt > 0.$$

□

12.2 A critical rank-one inverse spectrum

Put

$$\alpha_j = p_{2j-1}^{-1}, \quad \beta_j = p_{2j}^{-1},$$

so that $\alpha_1 > \beta_1 > \alpha_2 > \beta_2 > \cdots \downarrow 0$, and define

$$\Psi(z) := \prod_{j \geq 1} \frac{1 - z\beta_j}{1 - z\alpha_j}. \quad (129)$$

The product converges locally uniformly away from its poles because $\sum_j (\alpha_j - \beta_j) < \infty$.

Theorem 12.2 (Critical rank-one realization of prime interlacing). [A+E] *There are positive weights*

$$w_j = (\alpha_j - \beta_j) \prod_{k \neq j} \frac{1 - \beta_k/\alpha_j}{1 - \alpha_k/\alpha_j} \quad (130)$$

such that

$$\Psi(z) = 1 + z \sum_{j \geq 1} \frac{w_j}{1 - z\alpha_j}, \quad (131)$$

and

$$\sum_j w_j = \mathcal{P}_{\text{alt}}(1), \quad \sum_j \frac{w_j}{\alpha_j} = 1, \quad \sum_j \frac{w_j}{\alpha_j^2} = \infty. \quad (132)$$

Consequently Ψ is a meromorphic Pick function.

On $\ell^2(\mathbb{N})$ let

$$\mathbf{A}_o = \text{diag}(\alpha_1, \alpha_2, \dots), \quad v = (\sqrt{w_j})_{j \geq 1}, \quad \mathbf{A}_e = \mathbf{A}_o - v \otimes v.$$

Then $\mathbf{A}_o$ and $\mathbf{A}_e$ are positive compact injective operators and

$$\sigma(\mathbf{A}_o) = \{0\} \cup \{p_{2j-1}^{-1} : j \geq 1\}, \quad \sigma(\mathbf{A}_e) = \{0\} \cup \{p_{2j}^{-1} : j \geq 1\}. \quad (133)$$

For z away from the poles,

$$\Psi(z) = \det((I - z\mathbf{A}_e)(I - z\mathbf{A}_o)^{-1}). \quad (134)$$

Although neither operator is trace class, for every $\Re s > 0$ one has $\mathbf{A}_o^s - \mathbf{A}_e^s \in \mathcal{S}_1$ and

$$\mathrm{Tr}(\mathbf{A}_o^s - \mathbf{A}_e^s) = \sum_{j \geq 1} (\alpha_j^s - \beta_j^s) = \mathcal{P}_{\mathrm{alt}}(s), \quad (135)$$

where $x^s = e^{s \log x}$ for $x > 0$, $0^s = 0$, and the series is paired as displayed.

Finally, if $\mathbf{A}_{e,\tau} = \mathbf{A}_o - \tau v \otimes v$, then

$$\mathbf{A}_{e,\tau} \geq 0 \quad (0 \leq \tau \leq 1), \quad \mathbf{A}_{e,\tau} \text{ has exactly one negative eigenvalue} \quad (\tau > 1). \quad (136)$$

At $\tau = 1$, zero is a continuous-spectrum threshold, not an eigenvalue.

Proof. For a finite interlacing product, partial fractions give $\Psi_J(z) = 1 + z \sum_{j \leq J} w_{j,J} / (1 - z\alpha_j)$ with positive residues. Passing monotonically to the limit gives (130) and (131). The finite identities

$$\sum_{j \leq J} w_{j,J} = \sum_{j \leq J} (\alpha_j - \beta_j), \quad 1 - \sum_{j \leq J} \frac{w_{j,J}}{\alpha_j} = \prod_{j \leq J} \frac{\beta_j}{\alpha_j}$$

give the first two sums in (132); the last product tends to zero because the odd and even prime blocks both have positive lower density [8].

For $t > 0$ write

$$D_o(t) = \sum_j \int_{p_{2j-1}}^{p_{2j}} \frac{t \, du}{u(u+t)} = -\log \Psi(-t)$$

and define $D_e(t)$ using the complementary intervals $[p_{2j}, p_{2j+1}]$. Then $D_o(t) + D_e(t) = \log(1 + t/2)$, while both terms tend to infinity by the same positive-density theorem. Hence $t\Psi(-t) = 2t(t + 2)^{-1}e^{D_e(t)} \rightarrow \infty$. Since

$$\Psi(-t) = \sum_j \frac{w_j/\alpha_j}{1 + t\alpha_j},$$

monotone convergence proves the last sum in (132).

Cauchy–Schwarz now gives

$$|\langle u, v \rangle|^2 \leq \langle \mathbf{A}_o u, u \rangle \sum_j \frac{w_j}{\alpha_j} = \langle \mathbf{A}_o u, u \rangle,$$

so $\mathbf{A}_e \geq 0$. Equality would require u proportional to $\mathbf{A}_o^{-1}v$, but $\|\mathbf{A}_o^{-1}v\|^2 = \infty$, proving injectivity. The matrix determinant lemma gives (134); its interlacing zeros give (133). The same quadratic-form estimate proves positivity for $\tau \leq 1$, while finite truncations of $\mathbf{A}_o^{-1}v$ give one negative direction when $\tau > 1$; a rank-one subtraction gives at most one.

The Krein spectral-shift function is

$$\eta(x) = \mathbf{1}_{\cup_j (\beta_j, \alpha_j)}(x) = \xi(1/x) \quad \text{a.e.}$$

For $u > 0$ put

$$R_u = (\mathbf{A}_e + uI)^{-1} - (\mathbf{A}_o + uI)^{-1}, \quad d(u) = \Psi(-1/u).$$

Sherman–Morrison and (131) give

$$R_u = \frac{(\mathbf{A}_o + uI)^{-1}v \otimes (\mathbf{A}_o + uI)^{-1}v}{d(u)}, \quad \|R_u\|_1 = \frac{d'(u)}{d(u)}. \quad (137)$$

The estimate $T\Psi(-T) \rightarrow \infty$ proved above gives $d(u)/u \rightarrow \infty$ as $u \downarrow 0$, and hence $|\log d(u)| = O(|\log u|)$; also $\|R_u\|_1 = O(u^{-2})$ as $u \rightarrow \infty$. Integration by parts therefore shows

$$\int_0^\infty u^\sigma \|R_u\|_1 \, du < \infty \quad (0 < \sigma < 1).$$

For $0 < \Re s < 1$, the trace-norm convergent fractional-power formula

$$A_o^s - A_e^s = \frac{\sin \pi s}{\pi} \int_0^\infty u^s R_u \, du \quad (138)$$

proves membership in $\mathcal{S}_1$. Differentiating $d(u) = \prod_j (u + \beta_j)/(u + \alpha_j)$ and integrating termwise gives (135) in this strip. The identity

$$A_o^{2s} - A_e^{2s} = (A_o^s - A_e^s)A_o^s + A_e^s(A_o^s - A_e^s)$$

propagates trace-class membership to every $\Re s > 0$, and analytic continuation propagates the trace identity. This is the concrete resolvent form of the usual spectral-shift argument; compare [29, 32, 36]. $\square$

Corollary 12.3 (Prime subordinator and a strict moment inequality). *[A+E] $\Psi(-t)$ is the Laplace transform of an infinitely divisible random variable $X_{\mathbb{P}} \geq 0$ with Lévy measure*

$$\frac{1}{u} \sum_{j \geq 1} (e^{-p_{2j-1}u} - e^{-p_{2j}u}) \, du. \quad (139)$$

It has infinite activity, finite mean, and

$$\kappa_m(X_{\mathbb{P}}) = (m-1)! \mathcal{P}_{\text{alt}}(m) \quad (m \geq 1).$$

In particular,

$$4\mathcal{P}_{\text{alt}}(1)\mathcal{P}_{\text{alt}}(3) - 3\mathcal{P}_{\text{alt}}(2)^2 - \mathcal{P}_{\text{alt}}(1)^4 > 0. \quad (140)$$

Proof. Frullani's formula gives the Lévy–Khinchine representation of $\log \Psi(-t)$. The activity is infinite because $\sum_j \log(p_{2j}/p_{2j-1}) = \infty$, and the mean is $\mathcal{P}_{\text{alt}}(1)$. Differentiation at zero gives the cumulants. Finally, strict positivity of the 2×2 Hankel determinant for the infinite-support measure $\sum_j w_j \delta_{\alpha_j}$ reduces algebraically to (140); compare [41]. $\square$

12.3 The exact right edge in the half-plane of absolute convergence

Let $\mathfrak{P}(\sigma) = \sum_p p^{-\sigma}$ be the prime zeta function for $\sigma > 1$. Define σ_e by either of the equivalent equations

$$\sum_{p \geq 3} \left(\frac{2}{p}\right)^{\sigma_e} = 1, \quad \mathfrak{P}(\sigma_e) = 2^{1-\sigma_e}. \quad (141)$$

The solution is unique, and numerically

$$\sigma_e = 1.7795446535469941164458987869655 \dots \quad (142)$$

Let

$$Z_{>1} := \{\Re \rho : \mathcal{P}_{\text{alt}}(\rho) = 0, \Re \rho > 1\}.$$

Theorem 12.4 (Exact zero-projection edge). [L+E] *The closure of $Z_{>1}$ relative to $(1, \infty)$ is*

$$\overline{Z_{>1}}^{(1, \infty)} = (1, \sigma_e], \quad (143)$$

or, equivalently, its ordinary closure in $\mathbb{R}$ is

$$\overline{Z_{>1}}^{\mathbb{R}} = [1, \sigma_e]. \quad (144)$$

Moreover

$$\mathcal{P}_{\text{alt}}(s) \neq 0 \quad (\Re s \geq \sigma_e). \quad (145)$$

Thus $\sup Z_{>1} = \sigma_e$, but the supremum is not attained.

Proof. On $\Re s > 1$, (127) is absolutely convergent and uniformly almost periodic on every closed interior strip. Its frequencies $-\log p$ are linearly independent over $\mathbb{Q}$: after clearing denominators, any rational relation would contradict unique factorization.

The polygon criterion of Sepulcre–Vidal [23, Theorem 6] says that $\sigma > 1$ belongs to the closure of zero real projections exactly when every side $p^{-\sigma}$ is at most the sum of the remaining sides. Since $2^{-\sigma}$ is the largest, the only nonautomatic inequality is

$$2^{-\sigma} \leq \sum_{p \geq 3} p^{-\sigma},$$

which holds exactly for $1 < \sigma \leq \sigma_e$. This proves the two closure statements.

For $\sigma > \sigma_e$, the tail has modulus strictly smaller than $2^{-\sigma}$, so no zero is possible. Suppose a zero existed at $s = \sigma_e + it$. Equality would then hold in the triangle inequality, forcing all tail vectors to align. Comparing the terms for 3, 5 and for 5, 7 gives

$$t \log(5/3) \equiv \pi \pmod{2\pi}, \quad t \log(7/5) \equiv \pi \pmod{2\pi}.$$

Their logarithm ratio would be rational, which would force an impossible multiplicative relation between 5/3 and 7/5. Hence the boundary is also zero-free. $\square$

Theorem 12.5 (Linear order of zeros in the absolute-convergence strip). [L+E] *For every $1 < a < b < \sigma_e$, counting multiplicity,*

$$\#\{\rho : \mathcal{P}_{\text{alt}}(\rho) = 0, a < \Re \rho < b, 0 < \Im \rho < T\} \asymp_{a,b} T. \quad (146)$$

Proof. The closure theorem supplies an actual zero in every such open strip. Choose a small closed disk around one zero, with boundary free of zeros. Relatively dense almost periods approximate $\mathcal{P}_{\text{alt}}$ uniformly on the boundary; Rouché’s theorem therefore reproduces a zero in linearly many disjoint vertical translates, giving the lower bound.

For the upper bound, the compact vertical-translation hull consists of limits

$$g(s) = \sum_{k \geq 1} (-1)^{k+1} \omega_k p_k^{-s}, \quad |\omega_k| = 1.$$

At any fixed $B > \sigma_e$, first-term domination makes every hull element nonzero at B , so none is identically zero. Compactness, Hurwitz’s theorem, and a slightly enlarged unit-height rectangle give a uniform bound for the number of zeros in each unit box. Summing the boxes proves the upper bound. $\square$

The theorem determines linear order, not an asymptotic constant, and it does not say that every individual vertical line contains a zero.

12.4 Strong universality below the line of absolute convergence

Bayart and Kouroupis prove that $\sum_{n \geq 1} (-1)^n p_n^{-s} = -\mathcal{P}_{\text{alt}}(s)$ is strongly universal on $1/2 < \Re s < 1$ [3, Corollary 1.4]. The following consequence is therefore a literature application, not a new universality theorem.

Corollary 12.6 (Universality zeros). [L+E] *For every $1/2 < a < b < 1$,*

$$\#\{\rho : \mathcal{P}_{\text{alt}}(\rho) = 0, a < \Re \rho < b, 0 < \Im \rho < T\} \gg_{a,b} T. \quad (147)$$

Proof. Choose a closed disk K in the strip, centered at s_0 , and use the target $f(s) = s - s_0$. Strong universality gives a positive lower density of vertical shifts on which $\mathcal{P}_{\text{alt}}(s + i\tau)$ approximates f more closely on ∂K than $|f|$. Rouché's theorem puts a zero in every translated disk. A fixed zero can correspond to a τ -interval of length at most the disk diameter, so positive measure of good shifts requires linearly many distinct zeros. $\square$

12.5 The Jessen zero-measure limit

Theorem 12.7 (Jessen zero measure for $\mathcal{P}_{\text{alt}}$). [L+E] *For every $\sigma > 1/2$, the limit*

$$J(\sigma) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log |\mathcal{P}_{\text{alt}}(\sigma + it)| \, dt \quad (148)$$

exists. The function J is finite, continuous, and convex on $(1/2, \infty)$. Writing a zero as $\rho = \beta + i\gamma$ and counting its multiplicity m_ρ , set

$$\nu_T := \frac{1}{T} \sum_{\substack{\mathcal{P}_{\text{alt}}(\rho)=0 \\ 0 < \gamma < T, \beta > 1/2}} m_\rho \delta_\beta.$$

Then, as locally finite measures on $(1/2, \infty)$,

$$\nu_T \xrightarrow{v} \frac{1}{2\pi} J'', \quad (149)$$

where J'' is the distributional second derivative. If J is differentiable at a and b , with $1/2 < a < b$, then

$$N(a, b; T) := \sum_{\substack{\mathcal{P}_{\text{alt}}(\rho)=0 \\ a < \Re \rho < b, 0 < \Im \rho < T}} m_\rho$$

satisfies

$$N(a, b; T) = \frac{J'(b) - J'(a)}{2\pi} T + o(T). \quad (150)$$

The support of J'' in this half-line is exactly $(1/2, \sigma_e]$.

Proof. Let $P_N(s) = \sum_{k \leq N} (-1)^{k+1} p_k^{-s}$. These Dirichlet polynomials are analytic Bohr almost-periodic functions. For every $1/2 < a < b$, Landau's mean-square theorem [34, Part 24, Satz 37] applied to the tails gives

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_a^b |\mathcal{P}_{\text{alt}}(\sigma + it) - P_N(\sigma + it)|^2 \, d\sigma \, dt = \sum_{k > N} \int_a^b p_k^{-2\sigma} \, d\sigma,$$

and the right side tends to zero. This is precisely the area-mean approximation hypothesis in the generalized Jessen theorem of Borchsenius and Jessen [30, Theorem 1]; see also the ordinary almost-periodic theorem of Jessen and Tornehave [33]. Moreover, on every closed strip $1 < \alpha_0 \leq \Re s \leq \beta_0 < \infty$, absolute convergence gives $P_N \rightarrow \mathcal{P}_{\text{alt}}$ uniformly. Thus the Borchsenius–Jessen theorem applies with mean index $p = 2$ and yields existence, continuity and convexity of J and the interval formula (150) at differentiability endpoints. Since these endpoints are dense, the interval formula is equivalent to the vague convergence (149).

Given a nonempty open interval $I \Subset (1/2, 1)$, choose differentiability points $a < b$ of J in I . Then corollary 12.6 and (150) give $J''((a, b)) > 0$; the same argument with theorem 12.5 applies when $I \Subset (1, \sigma_e)$. Thus every such interval meets the support. The zero-free region (145) gives $J''((\sigma_e, \infty)) = 0$, intervals approaching σ_e from the left have positive mass, and every neighborhood of 1 meets one of the preceding regions. Hence the support relative to $(1/2, \infty)$ is exactly $(1/2, \sigma_e]$. $\square$

Remark 12.8. Below $\sigma = 1$ this is not a direct application of the ordinary Jessen–Tornehave Dirichlet-series theorem. Kronecker approximation gives $\sigma_u(\mathcal{P}_{\text{alt}}) = \sigma_a(\mathcal{P}_{\text{alt}}) = 1$; the extension to $\sigma > 1/2$ specifically uses the Borchsenius–Jessen area-mean theorem. Also, J'' is a Radon measure, not necessarily an ordinary density function.

12.6 Mellin–Barnes recovery of the parity prime constant

Theorem 12.9. [E] *For every $c > 0$,*

$$2C_{PP} = \frac{1}{\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s)(\log 2)^{-s} \mathcal{P}_{\text{alt}}(s) \, ds. \quad (151)$$

Proof. For $c > 1$, insert $x = p_k \log 2$ into Mellin inversion for e^{-x} and interchange the absolutely convergent sum and integral. This gives the same formula with C_{PP} and factor $1/(2\pi i)$. On closed strips in $\Re s > 0$, (128) gives $|\mathcal{P}_{\text{alt}}(s)| \ll (1 + |\Im s|)$, whereas $\Gamma(s)$ decays exponentially. Contour shifting is therefore valid to every $c > 0$. Multiplying by 2 gives (151). $\square$

13 The global parity frontier and Erdős’s alternating prime series

The fixed-lag and multiplicatively weighted results above should not be confused with unweighted cancellation of the prime-parity blocks. Put

$$D_r(x) := \#\{n \leq x : \pi(n) \equiv r \pmod{2}\}, \quad \mathcal{T}(x) := \sum_{n \leq x} (-1)^{\pi(n)} = D_0(x) - D_1(x). \quad (152)$$

Chung and Li proved the general unconditional estimate [8]

$$\liminf_{x \rightarrow \infty} \frac{1}{x} \#\{n \leq x : \pi(n) \equiv r \pmod{t}\} \geq \frac{1}{16t^2}. \quad (153)$$

At $t = 2$ this yields

$$\liminf \frac{D_0(x)}{x}, \quad \liminf \frac{D_1(x)}{x} \geq \frac{1}{64}, \quad \limsup \frac{|\mathcal{T}(x)|}{x} \leq \frac{31}{32}. \quad (154)$$

Alboiu independently proved positive proportion in every residue class [1]. Thus a simple Selberg argument giving, for example, $1/128$ would be valid but weaker than known work.

Proposition 13.1 (Thick excursions forced by large prime gaps). [L+E] *Let*

$$G(X) := \max_{p_{k+1} \leq X} (p_{k+1} - p_k).$$

Then

$$\max_{n \leq X} |\mathcal{T}(n)| \geq \frac{1}{2}G(X), \quad \#\{n \leq X : |\mathcal{T}(n)| > G(X)/4\} \geq \frac{1}{2}G(X) - O(1). \quad (155)$$

Consequently, for all sufficiently large X ,

$$\max_{n \leq X} |\mathcal{T}(n)| \gg \frac{\log X \log_2 X \log_4 X}{\log_3 X}. \quad (156)$$

Proof. On $[p_k, p_{k+1} - 1]$ the sign is constant and

$$\mathcal{T}(p_{k+1} - 1) - \mathcal{T}(p_k - 1) = (-1)^k g_k.$$

One endpoint therefore has modulus at least $g_k/2$. Among the g_k successive integer values traversed on the block, at most $g_k/2 + O(1)$ lie in $[-g_k/4, g_k/4]$. Maximize over the gaps below X and apply [12]. $\square$

The parity equidistribution problem

$$\mathcal{T}(x) = o(x) \quad (157)$$

remains open. It is equivalent to $D_0(x), D_1(x) \sim x/2$. Notice that (14) proves cancellation only after inserting μ or λ ; it does not remove those weights.

Theorem 13.2 (Punctured Fourier boundary). [A+E] *Let $\alpha > 0$. For every $|z| = 1$ with $z \neq 1$, the series*

$$\sum_{n \geq 2} \frac{(-1)^{\pi(n)} z^n}{n(\log n)^\alpha} \quad (158)$$

converges, and

$$\left| \sum_{n > M} \frac{(-1)^{\pi(n)} z^n}{n(\log n)^\alpha} \right| \ll_\alpha \frac{1}{|1 - z|(\log M)^\alpha}. \quad (159)$$

For $\alpha = 1$, every boundary frequency except $z = 1$ is therefore settled unconditionally.

Proof. Summation by parts first gives the exact identity

$$(1 - z) \sum_{n=1}^N \sigma_n z^n = z + 2 \sum_{p_k \leq N} (-1)^k z^{p_k} - \sigma_N z^{N+1}. \quad (160)$$

The prime number theorem, even in its elementary upper-bound form, implies

$$\sum_{n \leq N} \sigma_n z^n \ll \frac{N}{|1 - z| \log N}.$$

A second partial summation against $1/(n(\log n)^\alpha)$ proves (159). $\square$

Corollary 13.3 (The frequency $z = -1$). [E] *For every $X \geq 3$,*

$$\sum_{n \leq X} (-1)^{n+\pi(n)} \in \{-3, -2, -1\}. \quad (161)$$

Consequently $\sum_{n \geq 1} (-1)^{n+\pi(n)} n^{-s}$ has abscissa of convergence 0 and abscissa of absolute convergence 1.

Proof. For $m \geq 2$, the two terms at $2m - 1$ and $2m$ have the same value of π and opposite ordinary parity, so every completed pair cancels. The first two terms sum to -2 , and an unfinished pair changes this by ± 1 . $\square$

Tao [25] proved the exact equivalence between convergence of Erdős's series

$$\sum_{n \geq 1} \frac{(-1)^n n}{p_n} \quad (162)$$

and convergence of

$$\sum_{m \geq 2} \frac{(-1)^{\pi(m)}}{m \log m}. \quad (163)$$

The qualitative estimate (157) would not by itself settle this convergence: partial summation would leave a merely nonquantitative $o(1/(x \log x))$ integrand, which need not be integrable. A sufficient estimate is

$$\mathcal{T}(x) \ll \frac{x}{(\log \log x)^{1+\varepsilon}} \quad (164)$$

for some $\varepsilon > 0$; Tao displays the exponent 1.1 and obtains the needed conclusion under a strong Hardy–Littlewood prime-tuples hypothesis.

The honest hierarchy is therefore that published positive-proportion bounds are weaker than $\mathcal{T}(x) = o(x)$, which in turn is weaker than integrable quantitative cancellation such as (164). The next useful input must detect a signed first moment or short-interval parity correlation. Merely bounding parity-separated nonnegative second moments of the gaps does not force cancellation between odd- and even-indexed gap blocks.

14 Limitations, open problems, and audited status

14.1 What the paper does not prove

1. **Twin primes and phase tomography.** The curvature identity (38), the modulo-24 sensor (92), the threshold (45), and the phase decoder all re-encode finite prime-pair data. The decoder recovers $G_{r,X}(h)$ from supplied phase correlations, but gives no lower bound or recurrence theorem and does not recover the temporal order of gaps from one fixed- X correlation polynomial. In particular, they do not prove $H_* = 2$.
2. **Gilbreath's conjecture.** The affine inverse language, wheel criterion, corrected anchored right congruence, and transfer bounds describe obstruction words at fixed depth and height. They do not prove that actual prime data avoid every obstruction.
3. **Transcendence of C_{PP} .** We prove irrationality and base-2 non-normality, but neither the natural boundary of B nor formal transcendence over $\mathbb{F}_2(z)$ decides the arithmetic nature of $B(1/2)$.

4. **Algebraicity of the prime-gap continued fractions.** Irrationality exponent 2 is compatible with both algebraic and transcendental numbers. The Liouville statements concern $F_Q(r), F_H(r), F_N(r)$, not α_P or β .
5. **Euler’s constant and odd zeta values.** The Mellin boundary recovers $\gamma, \zeta(2), \zeta(3), \dots$ as Gumbel moments, but boundary limits do not preserve arithmetic type. No irrationality or transcendence result for γ , and no new individual transcendence result for an odd zeta value, follows.
6. **Global parity cancellation.** The Möbius–Liouville orthogonality theorem, the Green inversion (33), and the punctured Fourier theorem do not imply (157); even that open estimate would not by itself resolve Erdős’s series.
7. **Certified computation.** Numerical experiments and earlier “all pass” reports without source and certificate bundles are not used as theorem inputs.

14.2 Concrete next problems

1. Close the gap

$$\exp\left(\left(\frac{1}{16} - o(1)\right)(\log m)^2\right) \leq p_b(m) \leq \exp(Cm/\log m)$$

unconditionally. The CRT proof pinpoints the present bottleneck: Maynard’s theorem supplies $\asymp \log k$ primes among k candidates, whereas the conjectural optimum requires controlling a positive fraction or an arbitrary admissible subtuple.

2. Improve the logarithmic rate in [theorem 4.4](#), determine its sharp order, or obtain an effective theorem for general upper-Banach-sparse transition sets. The dynamical theorem [theorem 4.1](#) remains qualitative outside the prime-block setting.
3. Find genuinely quantitative input for the phase polynomials $\mathcal{F}_{X,h}(z)$. Fourier inversion is lossless decoding, not a recurrence theorem, and fixed- X data do not encode gap order.
4. Determine whether the finite-state Gilbreath quotient can be reduced polynomially in (H, r) while retaining actual-prime congruence data, or find an arithmetic invariant that separates surviving formal paths from genuine prime paths.
5. Determine algebraicity or transcendence of C_{PP} and of the three prime-gap continued fractions, and determine individual odd Gumbel cumulants. New functional equations or value-transcendence input would be required; natural boundaries and exact boundary recovery alone are insufficient.
6. In the Dirichlet problem, determine an explicit formula or finer regularity for the Jessen function J : in particular, decide whether J'' has atoms or an absolutely continuous component and describe zero statistics on individual vertical lines.
7. For the Erdős problem, first prove genuine cancellation $\mathcal{T}(x) = o(x)$, then strengthen it to an integrable rate such as (164). By (35), the first step is exactly a signed first-moment asymptotic for the all-lag correlation; a viable analytic target is unconditional decay of a short-interval Fourier coefficient of prime-gap index parity.

14.3 Claim-status summary

Topic	Strongest statement used here	Status
Factor complexity	$\log p_b(m) \geq (1/16 - o(1))(\log m)^2$ and $\log p_b(m) = O(m/\log m)$; conditional optimal order	A/L/C
Orthogonality	General qualitative sparse-transition theorem; uniform $O(x \log_3 x / \log_2 x)$ prime-block rate	L/E
Local word laws	Prime-rooted two-wall limit and reciprocal three-phase occurrence law	A/E
Finite diffraction	Exact recovery of $\pi(N)$ and a renormalized high-pass Haar limit	A/E
Correlation decoding	All-lag Green inversion, exact parity reconstruction, phase tomography, and H_* recovery	E
Gauss-continuant geometry	Common gap decoder, growth laws, irrationality exponent 2, and $\dim_H Z \leq 2/3$	A/E
Support series	Natural boundaries, hypertranscendence, and arithmetic value classifications	L/E
Gilbreath language	Exact anchored right congruence and finite transfer matrices; conjecture open	E
Dyadic numerators	Autonomous single-integer decoder; (N_n) is not P -recursive	L/E
Mellin boundary	$\Re s = 0$ is natural; all Gumbel moments are recovered from Z_N	A/E
Algebraic separation	$F_N(r)$ is algebraically independent from π ; relative $\sqrt{N}$ jet bound	L/E
Alternating prime spectrum	Critical rank-one model, exact right edge, and vague Jessen zero-measure limit	A/L/E
Global parity	Positive lower densities, thick excursions, and punctured Fourier convergence; $o(x)$ open	L/E
Balanced determinant	Exact trace field of transcendence degree 2 and rational-fibre formula	L/E
Prime-parity word algebra	Locally nilpotent nonnilpotent radical with $\dim \mathcal{J}_n = p_b(n) - 2$	E

A Quantitative truncated automaticity

Fix a base $q \geq 2$ and use the least-significant-digit-first q -adic convention in Dubbe's definition [11]. Let $A_{\mathbb{P}}^{(q)}(x)$ be the least number of states in an automaton recognizing primality correctly for all integers up to x , and let $A_b^{(q)}(x)$ be the analogous minimum for a DFAO producing b_n on that range.

Proposition A.1 (Transfer of Dubbe's lower bound). [L+E] *For every fixed q there is $C_q > 0$ such that*

$$A_b^{(q)}(x) \gg_q x^{1/2} \exp\left[-C_q(\log \log x)^2 \log \log \log x\right] \quad (165)$$

for all sufficiently large x .

Proof. Suppose a DFAO with S states produces b_n through the required range. Run one copy on n and a second copy behind a finite subtract-one/borrow transducer on $n-1$; take the XOR of the two outputs. By (2) the product machine recognizes primes and has $O_q(S^2)$ states. Therefore

$$A_{\mathbb{P}}^{(q)}(x) \ll_q (A_b^{(q)}(x+1))^2.$$

Dubbe's lower bound

$$A_{\mathbb{P}}^{(q)}(x) \geq x \exp \left[-c_q (\log \log x)^2 \log \log \log x \right]$$

gives (165) after taking square roots and changing the constant. $\square$

The $O_q(S^2)$ formulation is representation-safe. An exact $2S^2 + O(1)$ count requires a fixed high-order zero-padding convention so that subtracting one cannot create an unhandled terminal leading zero.

B Notation and reproducibility checklist

Symbol	Meaning
p_k, g_k	k -th prime and the consecutive gap $p_{k+1} - p_k$
χ_n, b_n, σ_n	prime indicator, cumulative prime parity, and its sign
$A(z), B(z)$	signed-prime difference series and cumulative-parity series
C_{PP}	$\sum_k (-1)^{k+1} 2^{-p_k}$; its digit word belongs to $2C_{PP}$
$C_X(h)$	prime-pair correlation weighted by prime-index difference parity
$D_X(h), \mathcal{F}_{X,h}(z)$	parity-overlap correlation and prime-rank phase polynomial
$\hat{\mu}_N, \hat{\nu}_N$	finite diffraction and its renormalized high-pass measure
x_k, δ_k	prime-time doubling point and its nearest-endpoint distance
z_k	corrected Gauss tail $G^k(\alpha_P) = [0; g_k, g_{k+1}, \dots]$
q_n, Q_m	full-gap convergent denominator and $Q_m = q_{2m}$
H_n	half-gap continuant denominator
N_n	odd numerator of $C_n = N_n/2^{p_n}$
F_Q, F_H, F_N	support series with exponents Q_m, H_n, N_n
Z_N, A_m	numerator Dirichlet series and its standard-Gumbel boundary moments
$\mathcal{P}_{\text{alt}}, \Psi$	alternating prime-index Dirichlet series and its perturbation determinant
$\mathbf{A}_o, \mathbf{A}_e$	compact operators carrying the odd- and even-index inverse-prime spectra
J, J''	Jessen function and its distributional zero-counting measure

All exact finite identities can be checked from a finite prime list:

1. build b_n and verify (2), (38), and (39);
2. iterate (51) and compare the three floors in (53);
3. iterate the half-gap recurrence and verify (66);
4. iterate (86) and verify the valuation (90);
5. implement the anchored Gilbreath state using (77) and (79), and verify the counterexample in remark 9.7.

Such computations are useful for detecting finite-identity errors, but they do not replace the limiting arguments or the cited analytic theorems.

C A balanced Gamma–Liouville determinant

Fix $r \in \mathbb{Q}$ with $0 < |r| < 1$ and set $L = L_r$. This appendix records a deliberately constructed bridge between this Liouville value and the Gamma trace tower. It is a consequence of the preceding

independence theorem, not a new general principle in operator theory. Put

$$\mathbf{C}_r = [L] \oplus \mathbf{A}_\Gamma \oplus (-\mathbf{A}_\Gamma).$$

Its nonexceptional eigenvalues are $1/n$ and $-1/n$, each with multiplicity five.

Theorem C.1 (Balanced trace tower and determinant). [\[E+L\]](#) $\mathbf{C}_r$ is compact, self-adjoint and Hilbert–Schmidt, but not trace class. Its balanced first trace and ordinary higher traces are

$$\begin{aligned} T_1 &:= \mathrm{Tr}_{\mathrm{pv}} \mathbf{C}_r := \lim_{N \rightarrow \infty} \mathrm{Tr}([L] \oplus \mathbf{A}_{\Gamma, N} \oplus (-\mathbf{A}_{\Gamma, N})) = L, \\ T_m &:= \mathrm{Tr} \mathbf{C}_r^m = \begin{cases} L^m, & m \text{ odd}, \\ L^m + 10\zeta(m), & m \text{ even} \end{cases} \quad (m \geq 2). \end{aligned} \tag{166}$$

Every T_m is transcendental,

$$\overline{\mathbb{Q}}(T_1, T_2, \dots) = \overline{\mathbb{Q}}(L, \pi^2), \quad \mathrm{trdeg}_{\overline{\mathbb{Q}}} = 2,$$

and two distinct traces are algebraically dependent exactly when both indices are odd.

The balanced regularized determinant is

$$\Delta_L(t) := e^{-tL} \det_2(I - t\mathbf{C}_r) = (1 - tL) \left(\frac{\sin \pi t}{\pi t} \right)^5. \tag{167}$$

Every nonconstant Taylor coefficient is transcendental. Its zero divisor consists of the nonzero integers, each with multiplicity five, and the simple Liouville zero $1/L$. If distinct $q_1, \dots, q_s \in \mathbb{Q} \setminus \mathbb{Z}$, then

$$\mathrm{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\Delta_L(q_1), \dots, \Delta_L(q_s)) = \min\{s, 2\}. \tag{168}$$

Proof. The spectral symmetry gives (166). Since L and π are algebraically independent by [theorem 11.9](#), all transcendence and field claims follow; for two even traces, the Jacobian of $x^a + c_a y^a$ and $x^b + c_b y^b$ is nonzero when $a \neq b$.

In the determinant product, exponential factors cancel between each $\pm 1/n$ pair and the exceptional eigenvalue contributes $(1 - tL)e^{tL}$. Euler’s sine product proves (167). If $(\sin x/x)^5 = \sum_{k \geq 0} q_k x^{2k}$, then $q_k \in \mathbb{Q}^\times$ (equivalently, these are nonzero even moments of a sum of five uniform variables, with alternating sign). Hence the even coefficients are $q_k \pi^{2k}$ and the odd ones are $-q_k L \pi^{2k}$.

For rational $q \notin \mathbb{Z}$, set $a_q = (\sin(\pi q)/q)^5 \in \overline{\mathbb{Q}}^\times$ and $x_q = \Delta_L(q)/a_q = \pi^{-5}(1 - qL)$. For distinct q_1, q_2 ,

$$\pi^{-5} = \frac{q_2 x_{q_1} - q_1 x_{q_2}}{q_2 - q_1}, \quad L = \frac{x_{q_1} - x_{q_2}}{q_2 x_{q_1} - q_1 x_{q_2}}.$$

Thus any pair recovers the independent generators and proves (168); one value is already transcendental. $\square$

D The radical of the prime-parity word algebra

Let K be a field and let $\mathrm{Fac}(b)$ be the factor language of the prime-parity word. Form the monomial algebra

$$\mathcal{A}_b := K\langle x_0, x_1 \rangle / \langle w : w \notin \mathrm{Fac}(b) \rangle, \tag{169}$$

and let $\mathcal{J}$ be the ideal spanned by surviving monomials containing both letters.

Theorem D.1 (Locally nilpotent complexity radical). [E]

$$\text{Jac}(\mathcal{A}_b) = \mathcal{J}, \quad \mathcal{A}_b/\mathcal{J} \cong K[x, y]/(xy). \quad (170)$$

The ideal $\mathcal{J}$ is locally nilpotent but not nilpotent, and its homogeneous pieces satisfy

$$\dim_K \mathcal{J}_n = p_b(n) - 2. \quad (171)$$

Thus all but two dimensions of the superpolynomial factor growth lie inside the Jacobson radical.

Proof. Each mixed surviving word contains a transition. Given finitely many generators of $\mathcal{J}$, any surviving product of m of them has at least m internal transitions and length at most Cm for a fixed C . This contradicts the upper Banach density zero of the prime transition set when m is large. Hence $\mathcal{J}$ is locally nilpotent and therefore lies in the Jacobson radical. It is not nilpotent, because arbitrarily long finite portions of b contain arbitrarily many prime transitions and can be split into arbitrarily many mixed factors.

Modulo $\mathcal{J}$, only pure powers remain, giving $K[x, y]/(xy)$, a semiprimitive algebra. Hence the radical is exactly $\mathcal{J}$. It remains to justify the two pure words in every degree. Choose distinct primes $q_j > |j|$ for $0 < |j| \leq n$ and solve $P \equiv -j \pmod{q_j}$. The resulting residue is coprime to $Q = \prod q_j$, so Dirichlet's theorem supplies a prime P in that progression; every $P+j$ with $0 < |j| \leq n$ is composite. The long constant blocks on the two sides of P have opposite bits, so both 0^n and 1^n occur. Exactly these two degree- n basis monomials lie outside $\mathcal{J}$, proving (171). For the general word-algebra framework, compare [28]. $\square$

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